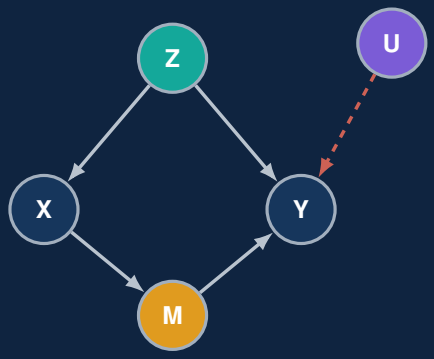


ADVANCED ARTIFICIAL INTELLIGENCE

Probabilistic Graphical Models and Causal Inference

Session 5

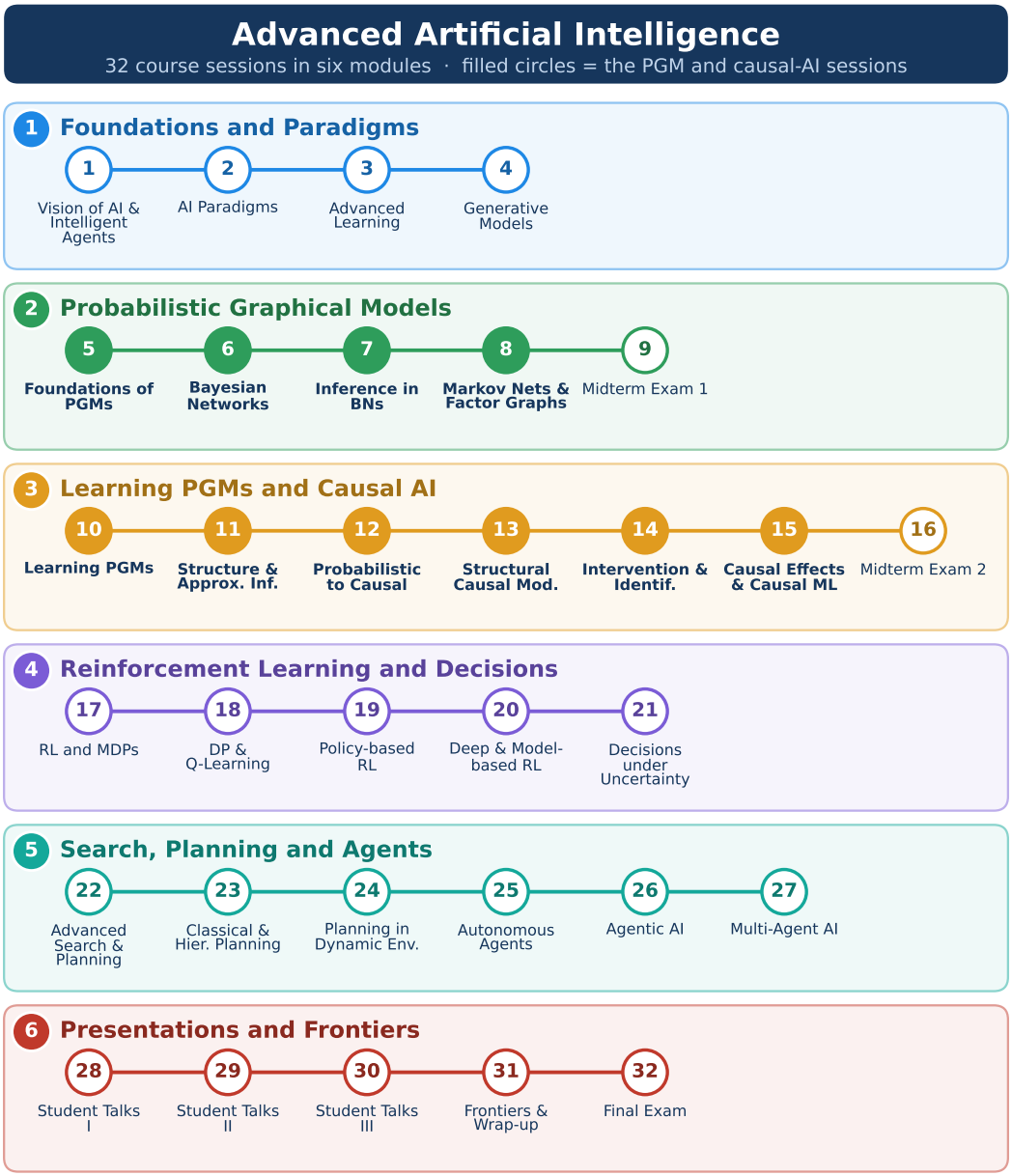
Foundations: Conditional Independence

Lecture Notes

Instructor: Prof. M. Arashi

About these notes

These notes cover Session 5 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, a "you are here" course map, the core theory organized in parts, a dedicated part on mathematical statistics, fully worked numerical and analytical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s05_examples.py` for the numbers, `python/s05_figures.py` for the figures), and `questions/s05_questions.tex` contains ten questions with complete answers.



The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions.

Contents

Session 5 Foundations: Conditional Independence **1**

5.1 Probability Review . . . 3
5.2 Why the Joint Distribution Alone Is Not Enough . . . 6
5.3 Independence and Conditional Independence . . . 7
5.4 The Algebra of Conditional Independence: Graphoid Axioms . . . 8
5.5 Worked Examples . . . 10
5.6 DAG Vocabulary . . . 14
5.7 From CI Statements to a DAG . . . 15
5.8 Conditional Independence and Information . . . 17
5.9 Gaussian Conditional Independence . . . 19
5.10 Sufficiency as Conditional Independence . . . 20
5.11 Testing Conditional Independence from Data . . . 21
5.12 Connections to AI, Agentic AI, RL and Multi-Agent Systems . . . 23
5.13 Summary . . . 27
5.14 Appendix: Derivations Omitted Above . . . 28

SESSION 5 ADVANCED ARTIFICIAL INTELLIGENCE

Foundations: Conditional Independence

Probability review, conditional independence, graphical representation, DAGs and factorization

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

PART I
Probability Review
Axioms, Bayes, total probability, expectation, tower rule and total variance.

PART II
Conditional Independence
Definition, factorized form, graphoid axioms, worked examples.

PART III
Graphs & Factorization
From CI statements to DAGs, topological order, parameter counting, I-maps.

PART IV
Mathematical Statistics
Mutual information, Gaussian CI, sufficiency, CI testing.

PART V
AI Connections
Belief states, the Markov property, routers, multi-agent fusion.

LEARNING OBJECTIVES

- State the probability tools used throughout the notes (Bayes' theorem, total probability, tower property, law of total variance) and apply them to small numerical problems.
- Define marginal and conditional independence, prove the factorized characterization and verify CI numerically from a joint table.
- State and prove the graphoid axioms (symmetry, decomposition, weak union, contraction, intersection) and explain why intersection needs positivity.
- Turn a list of CI statements into a DAG, use DAG vocabulary correctly, factorize along a topological order and count parameters with $\sum_i (k_i - 1) \prod_{j \in \mathrm{pa}(i)} k_j$.
- Characterize CI through conditional mutual information, Gaussian precision matrices and sufficient statistics, and test CI from data (G^2 and Fisher-z tests).
- Recognize CI as the assumption behind belief states, the Markov property of RL, POMDP observation models, naive-Bayes routers, VAEs and multi-agent sensor fusion.

Where we are. Course sessions 1–4 introduced rational agents, the symbolic/statistical/neural paradigms, generalization and deep generative models — all of which manipulate high-dimensional distributions. This session opens the PGM block; its organizing idea is *conditional independence* (CI), which turns an intractable joint distribution into a product of small, learnable pieces. Session 6 turns CI statements into Bayesian networks; Sessions 7–11 compute with and learn them;

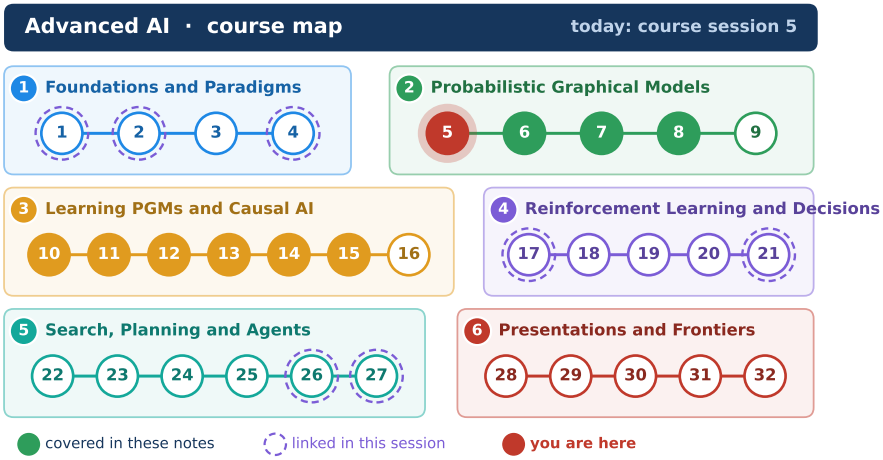


Figure 5.1. Where this session sits in the Advanced AI course. Dashed purple rings mark the course sessions that this chapter links to explicitly.

Sessions 12–15 give the arrows a causal meaning. Later, the same CI statements reappear as the Markov property of RL (course session 17), the POMDP observation model (21) and the belief state of LLM agents (26).

OPENING MOTIVATION

Suppose you want to build a full probabilistic model of $n = 50$ binary symptoms and diseases for a diagnostic system. The joint distribution $P(X_1, \ldots, X_{50})$ has $2^{50} - 1 \approx 1.13 \times 10^{15}$ free parameters — more numbers than any dataset could ever estimate and far more than any clinician could ever specify. Yet doctors reason about hundreds of symptoms every day. The resolution is not that the doctor secretly knows 2^{50} numbers; it is that most symptoms are *irrelevant to each other once you know the diagnosis*. This session makes that intuition precise. Conditional independence is the single idea that turns intractable joint distributions into tractable, learnable, and interpretable models — and it is the concept every remaining session in this course builds on.

Healthcare

Symptoms are roughly independent *given* the disease: one disease and 49 binary symptoms need $1 + 2 \cdot 49 = 99$ numbers instead of $2^{50} - 1 \approx 1.13 \times 10^{15}$.

Robotics

Sensor readings are CI given the pose; the next pose depends only on pose and action. This makes real-time filtering feasible.

LLM agents

If utterances are CI given the user's hidden intent, the agent's belief is updated by one multiplication per turn.

PART I

Probability Review

The language every later session speaks

PART I · PROBABILITY

5.1 Probability Review

5.1.1 Probability spaces and random variables

A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$. The *sample space* Ω collects all outcomes of an experiment; the *event space* $\mathcal{F}$ is a σ-algebra on Ω, i.e. a collection of subsets containing Ω and closed under complements and countable unions (for finite Ω, all subsets; the σ-algebra matters only for continuous spaces, where not every subset of $\mathbb{R}$ can be assigned a probability consistently); and $\mathbb{P} : \mathcal{F} \to [0,1]$ satisfies Kolmogorov's axioms:

$$\mathbb{P}(A) \geq 0, \qquad \mathbb{P}(\Omega) = 1, \qquad \mathbb{P}\Big(\bigcup_{k} A_k\Big) = \sum_{k} \mathbb{P}(A_k) \ \text{ for pairwise disjoint } A_1, A_2, \dots \tag{5.1}$$

A *random variable* is a measurable map $X : \Omega \to \mathcal{X}$; its distribution is $P_X(B) = \mathbb{P}(X^{-1}(B))$. We write $P(x)$ for $\mathbb{P}(X = x)$ when X is discrete (a probability mass function, pmf) and $p(x)$ for a density when X is continuous. For several variables, the *joint* distribution $P(x, y)$ determines the *marginals* $P(x) = \sum_y P(x, y)$ and, whenever $P(y) > 0$, the *conditionals*

$$P(x \mid y) = \frac{P(x, y)}{P(y)}. \tag{5.2}$$

Theorem 5.1 (Product rule, total probability and Bayes)**.** *For discrete random variables with the relevant probabilities positive:*

(i) ***Chain rule:*** $P(x_1, \dots, x_n) = \prod_{i=1}^{n} P(x_i \mid x_1, \dots, x_{i-1})$.

(ii) ***Law of total probability:*** $P(y) = \sum_x P(y \mid x) P(x)$.

(iii) ***Bayes' theorem:*** $P(x \mid y) = \dfrac{P(y \mid x) P(x)}{\sum_{x'} P(y \mid x') P(x')}$.

The same statements hold for densities with sums replaced by integrals.

Proof. (i) Apply (5.2) repeatedly: $P(x_1, \dots, x_n) = P(x_n \mid x_{<n}) P(x_{<n})$ and induct on n. (ii) The events $\{X = x\}$ partition Ω, so additivity in (5.1) gives $P(y) = \sum_x P(x, y) = \sum_x P(y \mid x) P(x)$. (iii) Write $P(x \mid y) = P(x, y)/P(y)$, expand the numerator by the product rule and the denominator by (ii). □

Worked Example 5.1 (Bayes' theorem for a screening test)**.** *Data (hypothetical).* $D \in \{0, 1\}$ indicates disease ($D{=}1$) or no disease ($D{=}0$); "+" and "−" denote a positive and a negative test result. The test has *sensitivity* $P(+ \mid D{=}1) = 0.95$ (probability that a diseased person tests positive), *specificity* $P(- \mid D{=}0) = 0.90$ (probability that a healthy person tests negative), and the *prevalence* is $P(D{=}1) = 0.02$. Find $P(D{=}1 \mid +)$, and then $P(D{=}1 \mid +, +)$ after a second, independent repetition of the test.

Solution.

Step 1. Complements. The theorem needs $P(+ \mid D{=}0)$, which is not given directly; it is the complement of the specificity (the *false-positive rate*): $P(+ \mid D{=}0) = 1 - P(- \mid D{=}0) = 1 - 0.90 = 0.10$. Likewise $P(D{=}0) = 1 - 0.02 = 0.98$.

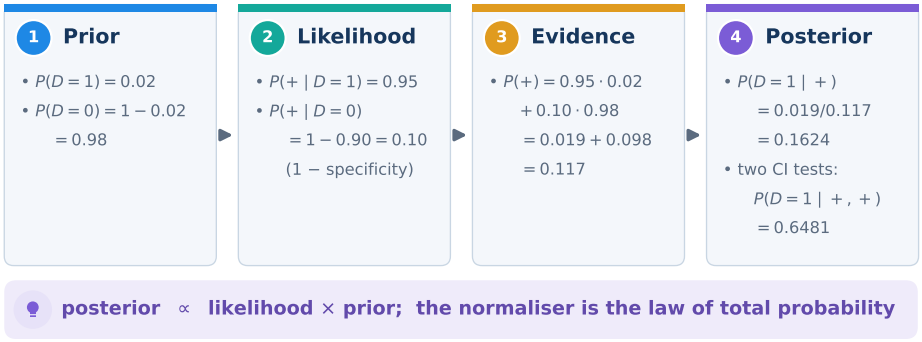


Figure 5.2. Bayes' theorem as a four-step flow, with the numbers of Worked Example 5.1. The false-positive rate $P(+ \mid D{=}0) = 1 - 0.90 = 0.10$ is one minus the specificity. The evidence (step 3) is the law of total probability; the second positive test is combined by multiplying likelihoods, which is only valid because the two tests are conditionally independent given disease status.

Step 2. Evidence by total probability (Theorem 5.1(ii)):

$$\begin{aligned} P(+) &= P(+ \mid D{=}1)P(D{=}1) + P(+ \mid D{=}0)P(D{=}0) \\ &= 0.95 \cdot 0.02 + 0.10 \cdot 0.98 = 0.019 + 0.098 = 0.117. \end{aligned}$$

Step 3. Posterior by Bayes (Theorem 5.1(iii)):

$$P(D{=}1 \mid +) = \frac{P(+ \mid D{=}1)P(D{=}1)}{P(+)} = \frac{0.019}{0.117} = 0.1624.$$

Of the 0.117 probability mass of positives, 0.098 comes from healthy people and only 0.019 from diseased ones: roughly one positive in six ($1/0.1624 \approx 6.2$) is a true case, because the 98% healthy majority produces most positives.

Step 4. A second positive test. Assume the two results T_1, T_2 are *conditionally independent given* D, $T_1 \perp\!\!\!\perp T_2 \mid D$ (Definition 5.2 below). Then the likelihoods multiply: $P(+,+ \mid D{=}1) = 0.95^2 = 0.9025$ and $P(+,+ \mid D{=}0) = 0.10^2 = 0.0100$. Hence

$$\begin{aligned} P(D{=}1 \mid +,+) &= \frac{P(+,+ \mid D{=}1)P(D{=}1)}{P(+,+ \mid D{=}1)P(D{=}1) + P(+,+ \mid D{=}0)P(D{=}0)} \\ &= \frac{0.9025 \cdot 0.02}{0.9025 \cdot 0.02 + 0.0100 \cdot 0.98} = \frac{0.01805}{0.01805 + 0.00980} \\ &= \frac{0.01805}{0.02785} = 0.6481. \end{aligned}$$

Check by sequential updating: use the first posterior 0.1624 as the new prior, $\frac{0.95 \cdot 0.1624}{0.95 \cdot 0.1624 + 0.10 \cdot 0.8376} = \frac{0.1543}{0.2380} = 0.6481$ — the same number, again because of CI.

Without the CI assumption we would need the joint table $P(T_1, T_2 \mid d)$ (three free numbers per value of d) instead of the single number $P(+ \mid d)$ — our first encounter with CI as a modelling assumption that makes computation possible.

PYTHON LAB · `s05_examples.py` · `ex_bayes_review()`

```
P(+|D=0) = 1 - 0.9 = 0.10   P(+) = 0.019 + 0.098 = 0.1170   P(D=1|+) =
0.0190/0.1170 = 0.1624
P(D=1|+,+) = 0.01805/(0.01805 + 0.00980) = 0.6481   sequential:
```

```
0.1543/0.2380 = 0.6481
```

5.1.2 Expectation, conditional expectation and the tower property

The expectation of $g(X)$ is $\mathbb{E}[g(X)] = \sum_x g(x)P(x)$ (or $\int g(x)p(x)\,dx$). The *conditional expectation* $\mathbb{E}[Y \mid X]$ is the random variable $m(X)$ where $m(x) = \mathbb{E}[Y \mid X{=}x] = \sum_y y\,P(y \mid x)$; it is the best predictor of Y from X in mean square. The conditional variance is $\mathrm{Var}(Y \mid X) = \mathbb{E}[(Y - \mathbb{E}[Y \mid X])^2 \mid X]$.

Proposition 5.1 (Tower property and law of total variance)**.** *For square-integrable Y:*

$$\mathbb{E}[Y] = \mathbb{E}\big[\mathbb{E}[Y \mid X]\big], \qquad \mathrm{Var}(Y) = \underbrace{\mathbb{E}\big[\mathrm{Var}(Y \mid X)\big]}_{\textit{within-group}} + \underbrace{\mathrm{Var}\big(\mathbb{E}[Y \mid X]\big)}_{\textit{between-group}}.$$

Proof. Tower (discrete case): $\mathbb{E}[\mathbb{E}[Y \mid X]] = \sum_x P(x) \sum_y y\,P(y \mid x) = \sum_{x,y} y\,P(x,y) = \mathbb{E}[Y]$. For the variance write $m(X) = \mathbb{E}[Y \mid X]$ and $Y - \mathbb{E}Y = (Y - m(X)) + (m(X) - \mathbb{E}Y)$. Squaring and taking expectations, $\mathrm{Var}(Y) = \mathbb{E}[(Y-m)^2] + \mathbb{E}[(m - \mathbb{E}Y)^2] + 2\mathbb{E}[(Y-m)(m-\mathbb{E}Y)]$. By the tower property the first term is $\mathbb{E}[\mathrm{Var}(Y \mid X)]$, the second is $\mathrm{Var}(m(X))$ since $\mathbb{E}[m(X)] = \mathbb{E}Y$, and the cross term is $2\mathbb{E}\big[(m(X) - \mathbb{E}Y)\,\mathbb{E}[Y - m(X) \mid X]\big] = 0$. □

Worked Example 5.2 (A latent class inflates the variance)**.** *Data (hypothetical).* A user of a recommender system rates $m = 10$ films; $R_j = 1$ if film j is rated highly. A latent class Z says whether the user is an enthusiast ($Z{=}1$) or not ($Z{=}0$), with $P(Z{=}1) = 0.3$ and hence $P(Z{=}0) = 0.7$. Given Z, the ten ratings are independent with $P(R_j{=}1 \mid Z) = p_Z$, where $p_1 = 0.9$ and $p_0 = 0.1$ (they are CI given Z). The number of high ratings $N = \sum_{j=1}^{10} R_j$ therefore satisfies $N \mid Z \sim \mathrm{Bin}(10, p_Z)$, so that

$$\mathbb{E}[N \mid Z] = 10\,p_Z, \qquad \mathrm{Var}(N \mid Z) = 10\,p_Z(1 - p_Z).$$

Here p_Z is itself a random variable: it equals 0.9 with probability 0.3 and 0.1 with probability 0.7. Compute $\mathbb{E}[N]$ and $\mathrm{Var}(N)$ with Proposition 5.1.

Solution.

Step 1. Mean (tower property). $\mathbb{E}[p_Z] = 0.3 \cdot 0.9 + 0.7 \cdot 0.1 = 0.27 + 0.07 = 0.34$, so $\mathbb{E}[N] = \mathbb{E}\big[\mathbb{E}[N \mid Z]\big] = 10\,\mathbb{E}[p_Z] = 10 \cdot 0.34 = 3.40$.

Step 2. Within-group term. $p_1(1-p_1) = 0.9 \cdot 0.1 = 0.09$ and $p_0(1-p_0) = 0.1 \cdot 0.9 = 0.09$, so

$$\mathbb{E}[\mathrm{Var}(N \mid Z)] = 10\,\mathbb{E}[p_Z(1-p_Z)] = 10\,(0.3 \cdot 0.09 + 0.7 \cdot 0.09) = 10 \cdot 0.09 = 0.90.$$

Step 3. Between-group term. $\mathrm{Var}(\mathbb{E}[N \mid Z]) = \mathrm{Var}(10\,p_Z) = 100\,\mathrm{Var}(p_Z) = 100\big(\mathbb{E}[p_Z^2] - (\mathbb{E}[p_Z])^2\big)$, with

$$\begin{aligned}
\mathbb{E}[p_Z^2] &= 0.3 \cdot 0.9^2 + 0.7 \cdot 0.1^2 = 0.3 \cdot 0.81 + 0.7 \cdot 0.01 \\
&= 0.243 + 0.007 = 0.250, \\
(\mathbb{E}[p_Z])^2 &= 0.34^2 = 0.1156, \\
\mathrm{Var}(p_Z) &= 0.250 - 0.1156 = 0.1344, \\
\mathrm{Var}(\mathbb{E}[N \mid Z]) &= 100 \cdot 0.1344 = 13.44.
\end{aligned}$$

Check: $\mathbb{E}[N \mid Z]$ takes the value 9 with probability 0.3 and 1 with probability 0.7, and a two-point variable has variance $0.3 \cdot 0.7 \cdot (9-1)^2 = 0.21 \cdot 64 = 13.44$.

Step 4. Law of total variance. $\mathrm{Var}(N) = 0.90 + 13.44 = 14.34$.

Comparison. If the ten ratings were fully independent, each high with probability 0.34, we would have $N \sim \mathrm{Bin}(10, 0.34)$ with variance $10 \cdot 0.34 \cdot 0.66 = 2.244$; the latent class makes the variance $14.34/2.244 = 6.39$ times larger. Over-dispersion of counts is a classic statistical fingerprint of a hidden common cause: the ratings are CI given Z, but strongly dependent marginally (indeed $\mathrm{Var}(p_Z) = 0.1344$ is exactly the covariance of two ratings, see Worked Example 5.6). A simulation confirms the result: drawing 400 000 users from this model (seed 5) gives sample mean 3.397 and sample variance 14.321.

PYTHON LAB · s05_examples.py · ex_total_variance()

```
E[p_Z] = 0.34, E[N] = 3.40   E[Var(N|Z)] = 10*0.09 = 0.9000   E[p_Z^2] =
0.243 + 0.007 = 0.250
Var(E[N|Z]) = 100*(0.250 - 0.1156) = 13.4400   Var(N) = 14.3400
Bin(10,0.34):  2.2440, ratio 6.39
Monte Carlo (400k draws, seed 5):  mean 3.397, var 14.321
```

PART I · PROBABILITY

5.2 Why the Joint Distribution Alone Is Not Enough

For discrete random variables $X_1, \ldots, X_n$, the chain rule of probability (Theorem 5.1) lets us write *any* joint distribution, with no assumptions at all, as

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P\big(x_i \mid x_1, \ldots, x_{i-1}\big). \tag{5.3}$$

This is always true, and it is almost always useless: the i-th factor is a table indexed by all $i-1$ preceding variables, so the total number of free parameters is still $2^n - 1$ for binary variables. The chain rule tells us a joint distribution *can* be factored; it says nothing about *when the factors are small*. Figure 5.3 makes the blow-up concrete: with only ten binary variables, a full joint table already needs $2^{10} - 1 = 1023$ parameters, while the chain $X_1 \to X_2 \to \cdots \to X_{10}$, in which each variable depends only on its predecessor, needs 1 number for $P(X_1{=}1)$ plus 2 numbers ($P(X_i{=}1 \mid x_{i-1})$ for $x_{i-1} = 0, 1$) for each of the other nine variables: $1 + 2 \cdot 9 = 19$.

The gap between these curves is entirely explained by conditional independence (CI) assumptions. This is the organizing idea of the whole PGM block: *a graphical model is nothing more than a compact way of writing down which conditional independencies hold, and a computationally efficient way of using them.* Everything from Session 6 onward is machinery built on top of this one idea.

PART II

Conditional Independence

Definition, factorization and the algebra of irrelevance

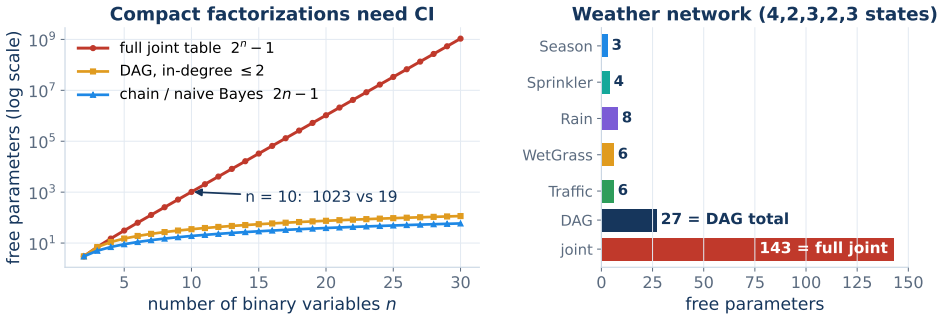


Figure 5.3. Left: free parameters required to specify a joint distribution over n binary variables (log scale): the full joint table ($2^n - 1$), a DAG in which every variable from the third on has two parents ($1 + 2 + 4(n-2)$), and a chain or naive-Bayes model ($1 + 2(n-1) = 2n - 1$). Right: the mixed-cardinality weather network of Worked Example 5.10 needs $3 + 4 + 8 + 6 + 6 = 27$ parameters instead of 143.

PART II · INDEPENDENCE

5.3 Independence and Conditional Independence

Definition 5.1 (Independence)**.** Random variables X and Y are *independent*, written $X \perp\!\!\!\perp Y$, if

$$P(x, y) = P(x)P(y) \quad \text{for all values } x, y,$$

equivalently $P(x \mid y) = P(x)$ whenever $P(y) > 0$: learning Y tells you nothing about X.

Independence is rare in practice — almost everything is at least weakly related to everything else. The far more useful and far more common structure is independence *given some context*.

Definition 5.2 (Conditional Independence)**.** Let X, Y, Z be (possibly vector-valued) random variables. X and Y are *conditionally independent given* Z, written $X \perp\!\!\!\perp Y \mid Z$, if

$$P(x, y \mid z) = P(x \mid z)\, P(y \mid z) \quad \text{for all } x, y, z \text{ with } P(z) > 0.$$

Equivalently, $P(x \mid y, z) = P(x \mid z)$ whenever $P(y, z) > 0$. For sets of variables, $\mathbf{X} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{Z}$ means the statement holds for the random vectors. Marginal independence is the special case $Z = \emptyset$.

Notation

CI gap (used in the Python checks). Multiplying the defining equation by $P(z)^2$ shows that $X \perp\!\!\!\perp Y \mid Z$ holds iff $P(x,y,z)P(z) = P(x,z)P(y,z)$ for all x, y, z (cells with $P(z) = 0$ satisfy it trivially). The function `ci_gap` in `s05_examples.py` returns $\Delta(X; Y \mid Z) = \max_{x,y,z} \left| P(x,y,z)P(z) - P(x,z)P(y,z) \right|$, which is 0 exactly when CI holds; for $Z = \emptyset$ it is $\max_{x,y} |P(x,y) - P(x)P(y)|$.

INTUITION

Think of Z as "using up" all of the statistical relationship between X and Y. Figure 5.4 shows this with overlapping regions representing shared information. In the middle panel, X and Y overlap directly: knowing X tells you something about Y that nothing else explains. On the right, X and Y still overlap, but the *entire* overlap sits inside Z — once you already know Z, there is nothing left of the X–Y relationship to learn. Part IV makes the picture

exact: the overlap is the (conditional) mutual information.

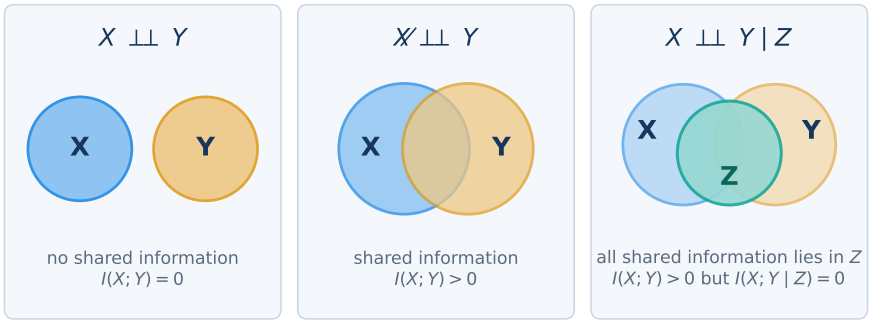


Figure 5.4. Information-diagram view. Left: independence, no shared information. Middle: dependence. Right: $X \perp\!\!\!\perp Y \mid Z$ — all shared information between X and Y routes through Z, so $I(X;Y) > 0$ while $I(X;Y \mid Z) = 0$ (Theorem 5.3).

An equivalent and often more convenient characterization avoids explicitly normalizing:

Proposition 5.2 (Factorized form of CI)**.** *$X \perp\!\!\!\perp Y \mid Z$ if and only if there exist functions g and h such that*

$$P(x, y, z) = g(x, z)\, h(y, z) \quad \text{for all } x, y, z.$$

Proof sketch. ($\Rightarrow$) Take $g(x,z) = P(x \mid z)P(z)$ and $h(y,z) = P(y \mid z)$; then $g(x,z)h(y,z) = P(x \mid z)P(y \mid z)P(z) = P(x, y \mid z)P(z) = P(x,y,z)$ using Definition 5.2. ($\Leftarrow$) Sum both sides over x to get $P(y,z) = h(y,z)\sum_x g(x,z)$, and similarly for $P(x,z)$; substitute back to recover Definition 5.2. Full algebra in Section 5.14. □

This factorized view is exactly why CI matters computationally: it says the three-way table $P(x,y,z)$ can be built from two smaller tables g and h instead of one large one — a preview of the factor-based computation in Session 7. It is also the template for three later results in this chapter: the DAG factorization (Theorem 5.2), the Gaussian precision-matrix criterion (Theorem 5.4) and the Fisher–Neyman factorization (Theorem 5.5).

COMMON PITFALL Neither $X \perp\!\!\!\perp Y$ nor $X \perp\!\!\!\perp Y \mid Z$ implies the other. A common cause can create marginal dependence that disappears given Z (Worked Example 5.6); conditioning on a common effect can create dependence that was absent marginally (Worked Example 5.7).

PART II · INDEPENDENCE

5.4 The Algebra of Conditional Independence: Graphoid Axioms

CI is not an arbitrary relation; it obeys algebraic rules that let us manipulate independence statements symbolically, the way we manipulate equations.

Proposition 5.3 (Properties of conditional independence)**.** *For disjoint sets of variables* X, Y, Z, W*:*

(i) ***Symmetry:*** $X \perp\!\!\!\perp Y \mid Z \iff Y \perp\!\!\!\perp X \mid Z$.

(ii) ***Decomposition:*** $X \perp\!\!\!\perp (Y, W) \mid Z \implies X \perp\!\!\!\perp Y \mid Z$.

(iii) ***Weak union:*** $X \perp\!\!\!\perp (Y, W) \mid Z \implies X \perp\!\!\!\perp Y \mid (Z, W)$.

These are straightforward consequences of Definition 5.2 (see Section 5.14 for derivations) but they matter practically: decomposition says a CI statement about a group of variables implies CI for any subgroup, while weak union says that once we know X is irrelevant to the pair (Y, W), it stays irrelevant to Y even after we additionally condition on W. In Session 6, an entire graphical calculus (d-separation) will exist precisely to let us read off consequences like these directly from a picture, without redoing this algebra every time. Two further rules complete the system.

Proposition 5.4 (Contraction and intersection)**.** *For disjoint* X, Y, Z, W*:*

(iv) ***Contraction:*** $X \perp\!\!\!\perp W \mid Z$ *and* $X \perp\!\!\!\perp Y \mid (Z, W) \implies X \perp\!\!\!\perp (Y, W) \mid Z$.

(v) ***Intersection:*** *if* P *is strictly positive,* $X \perp\!\!\!\perp Y \mid (Z, W)$ *and* $X \perp\!\!\!\perp W \mid (Z, Y) \implies X \perp\!\!\!\perp (Y, W) \mid Z$.

Proof. Contraction: by the two hypotheses $P(x \mid y, w, z) = P(x \mid w, z) = P(x \mid z)$ wherever defined, so $P(x, y, w \mid z) = P(x \mid y, w, z)P(y, w \mid z) = P(x \mid z)P(y, w \mid z)$. Intersection is proved in Section 5.14; the key step is that a function of (x, w, z) that equals a function of (x, y, z) for *all* (y, w) cannot depend on w — which needs every combination (y, w) to have positive probability. □

A relation satisfying (i)–(iv) is called a *semi-graphoid*; with (v) as well it is a *graphoid* (Pearl 2009; Koller & Friedman 2009, Sec. 2.1). Every probability distribution induces a semi-graphoid; strictly positive distributions induce graphoids. Graph separation also satisfies all five rules, which is the deep reason why graphs are a good language for CI.

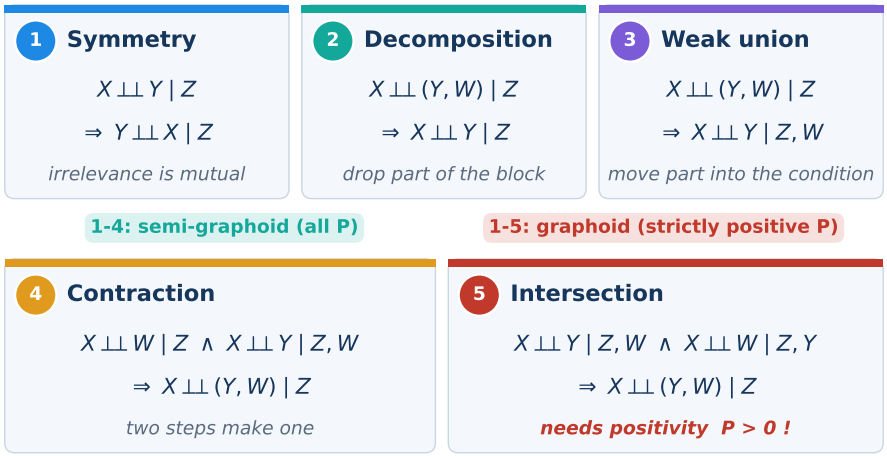


Figure 5.5. The five graphoid axioms as cards. Axioms 1–4 hold for every distribution; intersection requires strict positivity (Worked Example 5.3).

Worked Example 5.3 (Intersection fails without positivity)**.** Let $X = Y = W$ be three copies of one fair bit, so $P(X{=}0, Y{=}0, W{=}0) = P(X{=}1, Y{=}1, W{=}1) = \frac{1}{2}$ and the other

six cells are 0 (take $Z = \emptyset$). *Hypotheses hold.* Given $W = w$, both X and Y equal w, so $P(x, y \mid w) = \mathbb{1}\{x = w\}\mathbb{1}\{y = w\} = P(x \mid w)P(y \mid w)$ for all x, y: $X \perp\!\!\!\perp Y \mid W$. By the same argument with the roles of Y and W exchanged, $X \perp\!\!\!\perp W \mid Y$. *Conclusion fails.* $X \perp\!\!\!\perp (Y, W)$ would require $P(X{=}1, Y{=}1, W{=}1) = P(X{=}1)P(Y{=}1, W{=}1)$, but the left side is $\frac{1}{2}$ and the right side is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$; the CI gap is $\frac{1}{2} - \frac{1}{4} = 0.25$. In fact X determines (Y, W) completely: X carries one bit of uncertainty and none is left once (Y, W) is known, so $I(X; (Y, W)) = 1$ bit (mutual information is defined in Section 5.8). Intersection fails because the zero cells (e.g. $Y = 0, W = 1$) prevent us from "moving" between values of W while holding Y fixed. Deterministic relations — common in logic-based AI and in software agents — are exactly where positivity breaks.

PYTHON LAB · s05_examples.py · ex_intersection_counterexample()

```
gap X indep Y | W : 0.0   gap X indep W | Y : 0.0   gap X indep (Y,W) :
0.25, I = 1.0 bit
```

Table 5.1. The graphoid axioms at a glance.

Axiom	Statement	Needs $P > 0$?	One-line reason
Symmetry	$X \perp\!\!\!\perp Y \mid Z \Rightarrow Y \perp\!\!\!\perp X \mid Z$	no	product commutes
Decomposition	$X \perp\!\!\!\perp (Y, W) \mid Z \Rightarrow X \perp\!\!\!\perp Y \mid Z$	no	sum out w
Weak union	$X \perp\!\!\!\perp (Y, W) \mid Z \Rightarrow X \perp\!\!\!\perp Y \mid Z, W$	no	divide by $P(w \mid z)$
Contraction	$X \perp\!\!\!\perp W \mid Z \wedge X \perp\!\!\!\perp Y \mid Z, W \Rightarrow X \perp\!\!\!\perp (Y, W) \mid Z$	no	chain two equalities
Intersection	$X \perp\!\!\!\perp Y \mid Z, W \wedge X \perp\!\!\!\perp W \mid Z, Y \Rightarrow X \perp\!\!\!\perp (Y, W) \mid Z$	yes	needs all (y, w) cells

PART II · INDEPENDENCE

5.5 Worked Examples

Worked Example 5.4 (Symptoms given a diagnosis)**.** A clinic models three symptoms — fever S_1, cough S_2, fatigue S_3 — and a binary diagnosis $D \in \{\text{flu}, \text{no flu}\}$. Suppose the modeler assumes each symptom is generated independently once the diagnosis is fixed:

$$S_1 \perp\!\!\!\perp S_2 \perp\!\!\!\perp S_3 \mid D$$

(shorthand for every pair being CI given D, and jointly $P(s_1, s_2, s_3 \mid d) = \prod_{k=1}^{3} P(s_k \mid d)$; see Figure 5.6). This is the classical *naive Bayes* assumption. Without it, modeling $P(s_1, s_2, s_3 \mid d)$ requires $2^3 - 1 = 7$ parameters per value of d; with it, only 3 parameters per value of d are needed — one number $P(S_k{=}1 \mid d)$ per symptom. Including the prior $P(D{=}\text{flu})$, the model needs $1 + 2 \cdot 3 = 7$ numbers instead of $1 + 2 \cdot 7 = 15$. The assumption is almost certainly not perfectly true (a cough may make fatigue more likely even after conditioning on flu status), but it is often close enough to be highly useful, and it is learnable from a small dataset where the exact joint would not be.

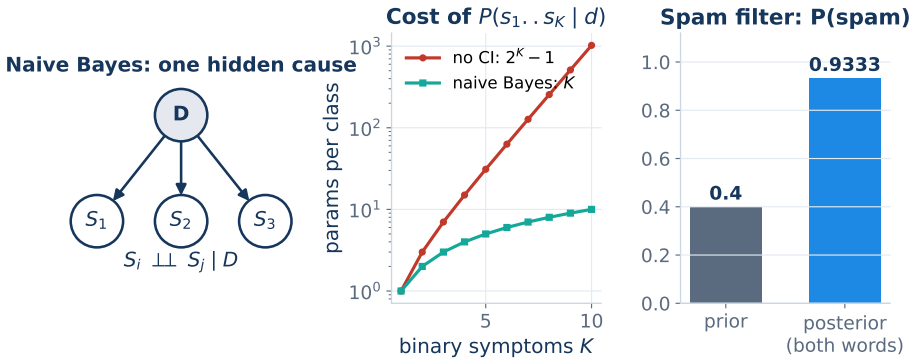


Figure 5.6. Left: a shared cause D explains all correlation among the symptoms; conditioning on D leaves the symptoms independent. Middle: parameters per class with and without the CI assumption. Right: prior and posterior spam probability in Worked Example 5.8.

Worked Example 5.5 (Verifying CI numerically)**.** Let $Z \in \{0,1\}$ be a fair coin, and let $X = Z \oplus N_1$, $Y = Z \oplus N_2$ where N_1, N_2 are independent fair coin flips (all independent of Z), and $\oplus$ is addition mod 2 ($0 \oplus 0 = 1 \oplus 1 = 0$, $0 \oplus 1 = 1 \oplus 0 = 1$). Since X and Y share the input Z one might expect them to be marginally dependent. Compute: $X = 1$ happens when $(Z, N_1) = (0,1)$ or $(1,0)$, so $P(X{=}1) = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$, and likewise $P(Y{=}1) = \frac{1}{2}$; moreover

$$P(X{=}1, Y{=}1) = P(Z{=}0)P(N_1{=}1)P(N_2{=}1) + P(Z{=}1)P(N_1{=}0)P(N_2{=}0) = \tfrac{1}{2} \cdot \tfrac{1}{4} + \tfrac{1}{2} \cdot \tfrac{1}{4} = \tfrac{1}{4},$$

which equals $P(X{=}1)P(Y{=}1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$: fair noise erases every marginal trace of Z, so this symmetric example happens to make them marginally independent too — a useful reminder that marginal independence and conditional independence are logically unrelated in general. Now condition on $Z = 0$: $X = N_1$ and $Y = N_2$, which are independent by construction, so $P(x, y \mid Z{=}0) = P(N_1{=}x)P(N_2{=}y) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = P(x \mid Z{=}0)P(y \mid Z{=}0)$ for every x, y; the same holds for $Z = 1$ (there $X = 1 - N_1$, $Y = 1 - N_2$). Hence $X \perp\!\!\!\perp Y \mid Z$ holds exactly, by Definition 5.2 — it is a statement that only becomes checkable once you fix Z.

PYTHON LAB · s05_examples.py · ex_xor_noise()

```
P(X=1) = P(Y=1) = 0.5   P(X=1,Y=1) = 0.25   P(X=1)P(Y=1) = 0.25   CI gap X
indep Y | Z : 0.0
```

Worked Example 5.6 (Marginal vs. conditional independence can point opposite ways)**.** Consider a recommendation system with $Z =$ "user is a movie enthusiast" (binary), $X =$ "user rated Movie A highly," $Y =$ "user rated Movie B highly." Suppose that, given enthusiast status, ratings for the two unrelated movies are independent: $X \perp\!\!\!\perp Y \mid Z$, with $P(X{=}1 \mid Z{=}1) = 0.9$, $P(X{=}1 \mid Z{=}0) = 0.1$, and identically for Y; enthusiasts make up 30% of users, so $P(Z{=}1) = 0.3$. Show that X and Y are *marginally dependent* despite being conditionally independent.

Solution. By the law of total probability,

$$P(X{=}1, Y{=}1) = \sum_z P(X{=}1 \mid z)P(Y{=}1 \mid z)P(z) = (0.9)^2(0.3) + (0.1)^2(0.7)$$
$$= 0.81 \cdot 0.3 + 0.01 \cdot 0.7 = 0.243 + 0.007 = 0.250,$$

where the first equality uses $X \perp\!\!\!\perp Y \mid Z$. Marginally $P(X{=}1) = 0.9(0.3) + 0.1(0.7) = 0.27 + 0.07 = 0.34$ (and the same for Y), so if X, Y were independent we would need $P(X{=}1, Y{=}1) = 0.34^2 = 0.1156$, far from the true value 0.250. Equivalently, $\mathrm{Cov}(X, Y) =$

$P(X{=}1,Y{=}1) - P(X{=}1)P(Y{=}1) = 0.250 - 0.1156 = 0.1344 > 0$, and learning $X = 1$ raises the probability of $Y = 1$ from 0.34 to $P(Y{=}1 \mid X{=}1) = 0.250/0.34 = 0.7353$. The unobserved enthusiasm variable Z *induces* an apparent correlation between two otherwise unrelated ratings — a small preview of the confounding phenomena that will become central in Session 12.

PYTHON LAB · s05_examples.py · ex_enthusiast()

```
P(X=1,Y=1) = 0.243 + 0.007 = 0.250   P(X=1) = 0.27 + 0.07 = 0.34; P(X=1)^2 =
0.1156
Cov(X,Y) = 0.1344   P(Y=1 | X=1) = 0.25/0.34 = 0.7353
```

Additional Worked Examples (Instructor Supplement)

Worked Example 5.7 (Pairwise independence does not imply mutual independence — and flips under conditioning)**.** Let X, Y be independent fair coin flips, and let $Z = X \oplus Y$ (addition mod 2). We show all three pairs are *marginally* independent, yet $X \not\perp\!\!\!\perp Y \mid Z$ — a second, sharper illustration (beyond the enthusiast example) of how marginal and conditional independence can point in opposite directions.

Marginal independence of every pair. By symmetry $P(X{=}1) = P(Y{=}1) = P(Z{=}1) = \frac{1}{2}$. Now $P(X{=}1, Z{=}1) = P(X{=}1, Y{=}0) = \frac{1}{4} = P(X{=}1)P(Z{=}1)$, so $X \perp\!\!\!\perp Z$; by an identical computation $Y \perp\!\!\!\perp Z$; and $X \perp\!\!\!\perp Y$ was assumed. So X, Y, Z are *pairwise* independent.

But not mutually independent. If X, Y, Z were mutually independent, we would need $P(X{=}1, Y{=}1, Z{=}1) = P(X{=}1)P(Y{=}1)P(Z{=}1) = \frac{1}{8}$. But $Z = X \oplus Y$, so $X{=}1, Y{=}1$ forces $Z{=}0$ with certainty: $P(X{=}1, Y{=}1, Z{=}1) = 0 \neq \frac{1}{8}$.

Conditioning on Z destroys the pairwise independence of X, Y. Given $Z = 0$, we know $Y = X$ exactly (since $X \oplus Y = 0 \iff X = Y$), so $P(X{=}1 \mid Y{=}1, Z{=}0) = 1 \neq P(X{=}1 \mid Z{=}0) = \frac{1}{2}$: knowing Y (given Z) tells you X with certainty. Hence $X \not\perp\!\!\!\perp Y \mid Z$, even though $X \perp\!\!\!\perp Y$ unconditionally. In terms of the CI gap: in the cell $(x, y, z) = (1, 1, 0)$, $P(1, 1, 0)P(Z{=}0) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$ while $P(X{=}1, Z{=}0)P(Y{=}1, Z{=}0) = \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{16}$, a gap of $\frac{1}{16} = 0.0625$. In information terms (Section 5.8) $I(X;Y) = 0$ but $I(X;Y \mid Z) = 1$ bit: given Z, X is still a fair bit, and Y then reveals it completely. This is the mirror image of Worked Example 5.6 (there, conditional independence held but marginal independence failed via a common cause; here, marginal independence holds for every pair but conditioning on the third variable makes two of them perfectly dependent) — together the two examples show that no relationship of implication holds in either direction between $X \perp\!\!\!\perp Y$ and $X \perp\!\!\!\perp Y \mid Z$ in general.

PYTHON LAB · s05_examples.py · ex_xor_pairwise()

```
pairwise CI gaps 0.0, 0.0, 0.0   P(X=1,Y=1,Z=1) = 0.0   gap X indep Y | Z =
0.0625
I(X;Y) = 0.0000 bits;  I(X;Y|Z) = 1.0000 bits
```

Worked Example 5.8 (A fully worked naive Bayes classification)**.** Return to the naive Bayes assumption from Worked Example 5.4 (symptoms conditionally independent given diagnosis), now with a concrete (hypothetical) email spam filter: class $C \in \{\text{spam}, \text{ham}\}$ (ham = legitimate email), and two word-presence indicators W_1, W_2 ($W_k = 1$ if word k occurs) with $W_1 \perp\!\!\!\perp W_2 \mid C$. Suppose $P(C{=}\text{spam}) = 0.4$, so $P(C{=}\text{ham}) = 1 - 0.4 = 0.6$, $P(W_1{=}1 \mid \text{spam}) = 0.7$,

$P(W_1{=}1 \mid \text{ham}) = 0.1$, $P(W_2{=}1 \mid \text{spam}) = 0.6$, $P(W_2{=}1 \mid \text{ham}) = 0.2$. An incoming email has both words present. By the CI assumption, $P(w_1, w_2 \mid c) = P(w_1 \mid c)P(w_2 \mid c)$, so

$$P(W_1{=}1, W_2{=}1 \mid \text{spam}) = 0.7 \times 0.6 = 0.42, \qquad P(W_1{=}1, W_2{=}1 \mid \text{ham}) = 0.1 \times 0.2 = 0.02.$$

By Bayes' rule, the unnormalized posteriors are $0.4 \times 0.42 = 0.168$ (spam) and $0.6 \times 0.02 = 0.012$ (ham), which sum to the evidence $P(W_1{=}1, W_2{=}1) = 0.168 + 0.012 = 0.180$, giving

$$P(\text{spam} \mid w_1, w_2) = \frac{0.168}{0.180} = \mathbf{0.9333}, \qquad P(\text{ham} \mid w_1, w_2) = \frac{0.012}{0.180} = 0.0667.$$

Without the CI assumption, classifying on two words would require the full joint table $P(w_1, w_2 \mid c)$ — $2^2 - 1 = 3$ free numbers per class instead of 2 — a small saving here, but one that becomes decisive with thousands of vocabulary words: the naive Bayes assumption is what makes text classification with a huge vocabulary statistically and computationally feasible at all.

PYTHON LAB · `s05_examples.py` · `ex_naive_bayes()`

```
spam:  likelihood 0.42, unnormalised 0.168, posterior 0.9333   ham:
likelihood 0.02, unnormalised 0.012, posterior 0.0667   evidence = 0.180
```

Worked Example 5.9 (Verifying weak union with explicit numbers)**.** The appendix proof of weak union is entirely algebraic; here is a concrete numerical check. Let $Z \in \{0, 1\}$ with $P(Z{=}0) = P(Z{=}1) = 0.5$, and suppose $X \perp\!\!\!\perp (Y, W) \mid Z$ (the weak-union hypothesis), with X's conditional distribution given by $P(X{=}1 \mid Z{=}0) = 0.3$, and (Y, W)'s *joint* conditional distribution given $Z{=}0$ specified directly by the table below. (We check the slice $Z{=}0$; the slice $Z{=}1$ works identically for any numbers.) Notably, Y and W are *correlated* given $Z{=}0$: $P(Y{=}1 \mid Z{=}0) = 0.4 + 0.1 = 0.5$ and $P(W{=}1 \mid Z{=}0) = 0.4 + 0.2 = 0.6$, so $P(Y{=}1 \mid Z{=}0)P(W{=}1 \mid Z{=}0) = 0.30 \neq 0.4 = P(Y{=}1, W{=}1 \mid Z{=}0)$. The table is

$$\begin{aligned} P(Y{=}1, W{=}1 \mid Z{=}0) &= 0.4, \quad P(Y{=}1, W{=}0 \mid Z{=}0) = 0.1, \\ P(Y{=}0, W{=}1 \mid Z{=}0) &= 0.2, \quad P(Y{=}0, W{=}0 \mid Z{=}0) = 0.3. \end{aligned}$$

Weak union claims $X \perp\!\!\!\perp Y \mid (Z, W)$ should now hold. Check it at $Z{=}0, W{=}1$: $P(W{=}1 \mid Z{=}0) = 0.4 + 0.2 = 0.6$, so $P(Y{=}1 \mid Z{=}0, W{=}1) = 0.4/0.6 = 2/3$. Since $X \perp\!\!\!\perp (Y, W) \mid Z$ by hypothesis, X does not depend on W given Z either (decomposition), so $P(X{=}1 \mid Z{=}0, W{=}1) = P(X{=}1 \mid Z{=}0) = 0.3$. Now compute the joint: $P(X{=}1, Y{=}1, W{=}1 \mid Z{=}0) = P(X{=}1 \mid Z{=}0)\, P(Y{=}1, W{=}1 \mid Z{=}0) = 0.3 \times 0.4 = 0.12$, so $P(X{=}1, Y{=}1 \mid Z{=}0, W{=}1) = 0.12/0.6 = 0.2$. Compare to the product $P(X{=}1 \mid Z{=}0, W{=}1)\, P(Y{=}1 \mid Z{=}0, W{=}1) = 0.3 \times \frac{2}{3} = 0.2$. They match exactly. The same computation at $Z{=}0, W{=}0$ gives $P(W{=}0 \mid Z{=}0) = 0.1 + 0.3 = 0.4$, $P(Y{=}1 \mid Z{=}0, W{=}0) = 0.1/0.4 = 0.25$ and $P(X{=}1, Y{=}1 \mid Z{=}0, W{=}0) = 0.3 \cdot 0.1/0.4 = 0.075 = 0.3 \times 0.25$. Since X and Y are binary, equality of the $(1, 1)$ cell with the product of the margins forces the other three cells of each 2×2 table to factorize as well (they are obtained by subtracting from the margins). This confirms $X \perp\!\!\!\perp Y \mid (Z, W)$ numerically, even though Y and W are themselves correlated given Z — weak union asks nothing about the relationship between Y and W, only that X stays irrelevant to Y once Z and W are both fixed.

PYTHON LAB · s05_examples.py · ex_weak_union()

```
W=1:  P(W=1|Z=0) = 0.6; P(Y=1|Z=0,W=1) = 0.6667; P(X=1,Y=1|Z=0,W=1) =
0.2000; product = 0.2000
W=0:  P(W=0|Z=0) = 0.4; P(Y=1|Z=0,W=0) = 0.2500; P(X=1,Y=1|Z=0,W=0) =
0.0750; product = 0.0750
```

PART III

Graphical Representation: DAGs and Factorization

From lists of CI statements to pictures

PART III · GRAPHS

5.6 DAG Vocabulary

Definition 5.3 (Directed acyclic graph)**.** A *DAG* $G = (V, E)$ is a finite set of nodes $V = \{X_1, \ldots, X_n\}$ with directed edges $E \subseteq V \times V$ and no directed cycle. For a node X: its *parents* $\mathrm{pa}(X) = \{U : U \to X\}$; *children* $\mathrm{ch}(X) = \{U : X \to U\}$; *ancestors* $\mathrm{an}(X)$: nodes with a directed path to X; *descendants* $\mathrm{de}(X)$: nodes reachable from X by a directed path; *non-descendants* $\mathrm{nd}(X) = V \setminus (\{X\} \cup \mathrm{de}(X))$. A *topological order* is an ordering of V in which every parent precedes its children.

Every DAG has at least one topological order: a finite DAG has a node without parents (otherwise following parent edges backwards would revisit a node and close a cycle); put it first, delete it and recurse. Parents are always non-descendants, $\mathrm{pa}(X) \subseteq \mathrm{nd}(X)$, since otherwise a directed cycle would exist.

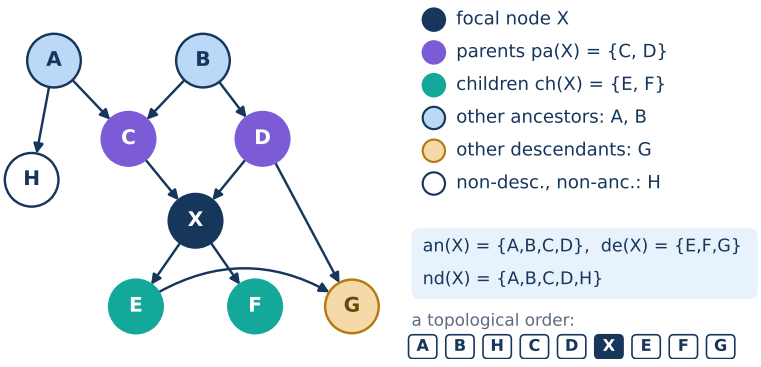


Figure 5.7. DAG vocabulary relative to a focal node X, and one topological order (there are several: e.g. H may be placed anywhere after A).

PART III · GRAPHS

5.7 From CI Statements to a DAG

Pick an ordering $X_1, \dots, X_n$ of the variables. Suppose that for each i we can find a (small) subset $\mathrm{pa}(X_i) \subseteq \{X_1, \dots, X_{i-1}\}$ such that

$$X_i \perp\!\!\!\perp \{X_1, \dots, X_{i-1}\} \setminus \mathrm{pa}(X_i) \mid \mathrm{pa}(X_i). \tag{5.4}$$

Then each factor of the chain rule (5.3) simplifies, $P(x_i \mid x_1, \dots, x_{i-1}) = P(x_i \mid \mathrm{pa}(x_i))$, and drawing an arrow from each member of $\mathrm{pa}(X_i)$ to X_i produces a DAG. This is the whole recipe for a Bayesian network; Session 6 develops it in full.

Definition 5.4 (Local Markov property; I-map)**.** Let $\mathcal{I}(P)$ be the set of CI statements that hold in P. A DAG G satisfies the *local Markov property* for P if

$$\mathcal{I}_\ell(G) = \big\{ X \perp\!\!\!\perp \mathrm{nd}(X) \setminus \mathrm{pa}(X) \mid \mathrm{pa}(X) \;:\; X \in V \big\} \subseteq \mathcal{I}(P).$$

We then call G an *I-map* (independence map) of P. It is a *minimal* I-map if removing any edge destroys this property.

Theorem 5.2 (Factorization along a topological order)**.** *Let G be a DAG over $X_1, \dots, X_n$. Then G is an I-map of P if and only if P factorizes over G:*

$$P(x_1, \dots, x_n) = \prod_{i=1}^{n} P\big(x_i \mid \mathrm{pa}(x_i)\big). \tag{5.5}$$

Proof. ($\Rightarrow$) Relabel so that $X_1, \dots, X_n$ is a topological order (it exists, see above). The predecessors $\{X_1, \dots, X_{i-1}\}$ contain $\mathrm{pa}(X_i)$ and are non-descendants of X_i. The local Markov property gives $X_i \perp\!\!\!\perp \mathrm{nd}(X_i) \setminus \mathrm{pa}(X_i) \mid \mathrm{pa}(X_i)$, and by decomposition (Proposition 5.3) $X_i \perp\!\!\!\perp \{X_1, \dots, X_{i-1}\} \setminus \mathrm{pa}(X_i) \mid \mathrm{pa}(X_i)$, i.e. (5.4). Substituting into the chain rule (5.3) gives (5.5). ($\Leftarrow$) Proved in Section 5.14 by summing out the descendants of X_i leaf by leaf. □

5.7.1 Parameter counting

If X_i has k_i states, the CPT $P(X_i \mid \mathrm{pa}(X_i))$ has one probability vector (with $k_i - 1$ free entries) for each of the $\prod_{j \in \mathrm{pa}(i)} k_j$ parent configurations. Hence

$$\#\mathrm{params}(G) = \sum_{i=1}^{n} (k_i - 1) \prod_{j \in \mathrm{pa}(i)} k_j \qquad \text{versus} \qquad \prod_{i=1}^{n} k_i - 1 \ \text{ for the full joint.} \tag{5.6}$$

For binary variables and in-degree at most d the BN needs at most $n 2^d$ numbers: linear rather than exponential in n.

Worked Example 5.10 (Counting parameters with mixed cardinalities)**.** A hypothetical weather network has Season (4 states) → Sprinkler (2), Season → Rain (3: none/light/heavy), (Sprinkler, Rain) → WetGrass (2), Rain → Traffic (3: light/medium/heavy). Formula (5.6) gives, node by node (Table 5.2): Season $(4-1)\cdot 1 = 3$; Sprinkler $(2-1)\cdot 4 = 4$; Rain $(3-1)\cdot 4 = 8$; WetGrass $(2-1)\cdot(2\cdot 3) = 6$; Traffic $(3-1)\cdot 3 = 6$. In total $3+4+8+6+6 = 27$

free parameters, while the full joint over $4 \cdot 2 \cdot 3 \cdot 2 \cdot 3 = 144$ configurations needs $144 - 1 = 143$ — a factor of $143/27 = 5.30$ even for five variables.

Table 5.2. Parameter count of the weather network (Worked Example 5.10).

Node	k_i	Parents	$\prod_{j \in \mathrm{pa}} k_j$	$(k_i - 1) \prod k_j$
Season	4	–	1	3
Sprinkler	2	Season	4	4
Rain	3	Season	4	8
WetGrass	2	Sprinkler, Rain	6	6
Traffic	3	Rain	3	6
Total				**27** (full joint: 143)

PYTHON LAB · s05_examples.py · ex_param_count()

```
Season (4-1)x1=3, Sprinkler (2-1)x4=4, Rain (3-1)x4=8, WetGrass (2-1)x6=6,
Traffic (3-1)x3=6
BN total = 27   full joint = 144 - 1 = 143   ratio 5.30
```

5.7.2 A first look at I-maps: the ordering matters

For a fixed joint P, the recipe (5.4) with the *smallest* admissible parent sets yields a minimal I-map, but which I-map we get depends on the ordering. Under positivity the smallest parent set for a given ordering is unique: if two different sets U_1, U_2 of the same minimal size both satisfied (5.4), their intersection $U_1 \cap U_2$ would satisfy it too (a five-line argument using weak union, intersection, decomposition and contraction, given in Section 5.14), contradicting minimality.

Worked Example 5.11 (Minimal I-maps of the sprinkler distribution for all 24 orderings)**.** Take the binary sprinkler network (Cloudy, Sprinkler, Rain, WetGrass) $C \to S$, $C \to R$, $(S, R) \to W$ with illustrative CPTs $P(C{=}1) = 0.5$; $P(S{=}1 \mid C{=}0) = 0.5$, $P(S{=}1 \mid C{=}1) = 0.1$; $P(R{=}1 \mid C{=}0) = 0.2$, $P(R{=}1 \mid C{=}1) = 0.8$; $P(W{=}1 \mid s, r) = 0.05, 0.9, 0.9, 0.99$ for $(s, r) = (0,0), (1,0), (0,1), (1,1)$. The script `ex_imap_orders()` computes the joint (16 cells) by (5.5); then, for every ordering and for each variable X_i in turn, it tries subsets U of the predecessors in order of increasing size and takes the first one with CI gap $\Delta\big(X_i; \text{predecessors} \setminus U \mid U\big) < 10^{-10}$, i.e. the smallest parent set satisfying (5.4). Parameters are counted with (5.6); a binary node with $|\mathrm{pa}|$ binary parents costs $2^{|\mathrm{pa}|}$.

- **Causal ordering** C, S, R, W recovers the generating graph: 4 edges and $1+2+2+4 = 9$ parameters (for C, S, R, W respectively).
- **Ordering** W, R, S, C gives $\mathrm{pa}(R) = \{W\}$, $\mathrm{pa}(S) = \{R, W\}$, $\mathrm{pa}(C) = \{S, R\}$: 5 edges and $1 + 2 + 4 + 4 = 11$ parameters. S needs *both* predecessors: $S \not\perp\!\!\!\perp R \mid W$ (gap 0.0684) and $S \not\perp\!\!\!\perp W \mid R$ (gap 0.0518). The first is "explaining away" (Session 6): conditioning on the common effect W couples its causes, and the coupling survives even when the common cause is known, since $S \perp\!\!\!\perp R \mid C$ (gap 0) but $S \not\perp\!\!\!\perp R \mid C, W$ (gap 0.0076). Finally $C \perp\!\!\!\perp W \mid S, R$ (gap $< 10^{-18}$), so $\mathrm{pa}(C) = \{S, R\}$.
- **Ordering** W, C, S, R even yields a complete DAG: $\mathrm{pa}(C) = \{W\}$, $\mathrm{pa}(S) = \{C, W\}$, $\mathrm{pa}(R) = \{C, S, W\}$, i.e. 6 edges and $1 + 2 + 4 + 8 = 15$ parameters — no saving at all

over the full joint ($2^4 - 1 = 15$).

Across all 24 orderings: 4 give 9 parameters, 8 give 11 and 12 give 15 (Figure 5.8).

PYTHON LAB · s05_examples.py · ex_imap_orders()

```
order CSRW: params 9, edges 4   order WRSC: params 11, edges 5   order WCSR:
params 15, edges 6
counts over 24 orderings:  {9:  4, 11:  8, 15:  12}   gaps:  S|R given W
0.0684, S|W given R 0.0518
S|R given C 0.00e+00, S|R given C,W 0.0076, C|W given S,R 4.34e-19
```

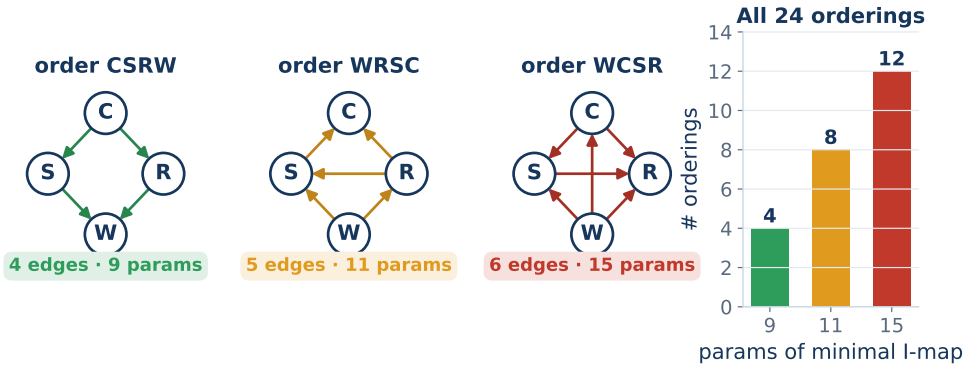


Figure 5.8. Minimal I-maps of the same sprinkler distribution for three orderings, and the distribution of their parameter counts over all 24 orderings. Orderings that follow cause before effect give the sparsest graphs.

Every distribution has many I-maps. Sparse ones are found by ordering causes before effects — the first hint that causal knowledge (Sessions 12–15) makes graphical models compact.

PART IV

Mathematical Statistics of Conditional Independence

Information, Gaussians, sufficiency and testing

PART IV · STATISTICS

5.8 Conditional Independence and Information

MATHEMATICAL STATISTICS · CI is zero conditional mutual information

For discrete X, Y, Z define the conditional mutual information

$$I(X;Y \mid Z) = \sum_{x,y,z} P(x,y,z) \log \frac{P(x,y \mid z)}{P(x \mid z)P(y \mid z)}$$
$$= \mathbb{E}_Z\Big[\mathrm{KL}\big(P(X,Y \mid Z) \,\big\|\, P(X \mid Z)P(Y \mid Z)\big)\Big].$$

It measures how far the conditional joint is from the CI product. With the conditional entropy $H(X \mid Z) = -\sum_{x,z} P(x,z) \log P(x \mid z)$ (and $H(X \mid Y,Z)$ defined in the same way), equivalently $I(X;Y \mid Z) = H(X \mid Z) - H(X \mid Y,Z)$: the reduction in uncertainty about X obtained from Y once Z is known. With $\log_2$ the unit is the *bit*, with the natural log the *nat* (1 bit $= \ln 2 = 0.6931$ nats). For $Z = \emptyset$ we obtain the mutual information $I(X;Y) = \sum_{x,y} P(x,y) \log \frac{P(x,y)}{P(x)P(y)}$.

Theorem 5.3 (Information characterization of CI)**.** *$I(X;Y \mid Z) \geq 0$, with equality if and only if $X \perp\!\!\!\perp Y \mid Z$.*

Proof. Fix z with $P(z) > 0$ and write $p(x,y) = P(x,y \mid z)$, $q(x,y) = P(x \mid z)P(y \mid z)$; both are pmfs. On the support of p, by Jensen's inequality for the strictly concave log,

$$-\mathrm{KL}(p\|q) = \sum_{x,y:\, p>0} p(x,y) \log \frac{q(x,y)}{p(x,y)} \;\leq\; \log \sum_{x,y:\, p>0} q(x,y) \;\leq\; \log 1 = 0$$

(Gibbs' inequality). Equality in Jensen forces q/p to be constant on the support of p, and equality in the second step forces q to put all its mass on that support; together $q = p$. Averaging over z with weights $P(z) > 0$: $I(X;Y \mid Z) = \sum_z P(z)\mathrm{KL}(p_z\|q_z) \geq 0$, and $I = 0$ iff $p_z = q_z$ for every z with $P(z) > 0$, which is Definition 5.2. □

Worked Example 5.12 (Mutual information of the enthusiast and XOR models)**.** *Enthusiast model* (Worked Example 5.6). The 2×2 table of (X,Y) follows from $P(X{=}1,Y{=}1) = 0.250$ and $P(X{=}1) = P(Y{=}1) = 0.34$: $P(1,0) = P(0,1) = 0.34 - 0.250 = 0.09$ and $P(0,0) = 1 - 0.250 - 0.09 - 0.09 = 0.57$. With $P(X{=}1)P(Y{=}1) = 0.1156$, $P(X{=}1)P(Y{=}0) = 0.34 \cdot 0.66 = 0.2244$ and $P(X{=}0)P(Y{=}0) = 0.66^2 = 0.4356$, the four terms of $I(X;Y)$ are

$$\underbrace{0.25 \log_2 \tfrac{0.25}{0.1156}}_{+0.2782} + 2 \cdot \underbrace{0.09 \log_2 \tfrac{0.09}{0.2244}}_{-0.1186} + \underbrace{0.57 \log_2 \tfrac{0.57}{0.4356}}_{+0.2211} = 0.2621 \text{ bits}$$

$(= 0.2621 \cdot \ln 2 = 0.1817$ nats). Negative terms are allowed; only the total is non-negative. Given Z, every ratio $P(x,y \mid z)/(P(x \mid z)P(y \mid z))$ equals 1 by CI, so every term of $I(X;Y \mid Z)$ is $\log 1 = 0$; the script returns 9.24×10^{-17}, which is floating-point rounding. Conditioning removes all shared information. *XOR model* (Worked Example 5.7). It is the other way round: $X \perp\!\!\!\perp Y$ gives $I(X;Y) = 0$; given $Z = z$, the two cells with $x \oplus y = z$ have $P(x,y \mid z) = \frac{1}{2}$ while $P(x \mid z)P(y \mid z) = \frac{1}{4}$, so each term is $\frac{1}{2}\log_2 2$ and $I(X;Y \mid Z) = \sum_z P(z)\left(\frac{1}{2} + \frac{1}{2}\right) = 1$ bit. Figure 5.9 maps $I(X;Y)$ over the whole family $P(X{=}1 \mid Z{=}1) = P(Y{=}1 \mid Z{=}1) = a$, $P(X{=}1 \mid Z{=}0) = P(Y{=}1 \mid Z{=}0) = b$ (with $P(Z{=}1) = 0.3$), and shows that if a direct link $X \to Y$ of strength ε is added (with probability ε, Y copies X; otherwise Y is drawn from $P(Y \mid Z)$ as before), $I(X;Y \mid Z)$ becomes positive for every $\varepsilon > 0$.

PYTHON LAB · s05_examples.py · ex_mutual_information()

```
cells (1,1),(1,0),(0,1),(0,0):  terms +0.2782, -0.1186, -0.1186, +0.2211
bits
enthusiast:  I(X;Y) = 0.2621 bits, I(X;Y|Z) = 9.24e-17   I(X;Y) = 0.1817
nats
XOR: I(X;Y) = 0.0000 bits, I(X;Y|Z) = 1.0000 bits
```

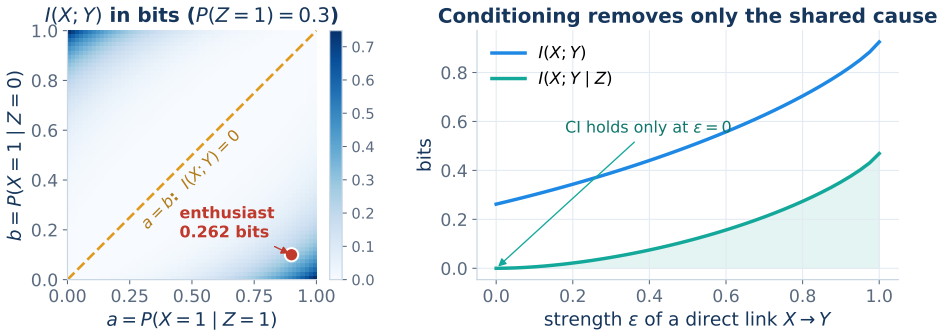


Figure 5.9. Left: $I(X;Y)$ for two ratings that are CI given a latent class Z, as a function of the class-conditional probabilities; it vanishes only on the diagonal $a = b$, where Z is irrelevant. Right: adding a direct link $X \to Y$ makes $I(X;Y \mid Z) > 0$.

PART IV · STATISTICS

5.9 Gaussian Conditional Independence

MATHEMATICAL STATISTICS · Precision matrix and partial correlation

Let $X \sim N_d(\mu, \Sigma)$ with $\Sigma \succ 0$ and *precision matrix* $\Theta = \Sigma^{-1}$. The *partial correlation* of X_i, X_j given all remaining variables is

$$\rho_{ij\cdot\text{rest}} = -\frac{\Theta_{ij}}{\sqrt{\Theta_{ii}\Theta_{jj}}}, \qquad \text{and for three variables} \qquad \rho_{13\cdot 2} = \frac{\rho_{13} - \rho_{12}\rho_{23}}{\sqrt{(1-\rho_{12}^2)(1-\rho_{23}^2)}}.$$

Zeros of Σ encode marginal independence; zeros of Θ encode conditional independence.

Theorem 5.4 (Gaussian CI and the precision matrix)**.** *For $X \sim N_d(\mu, \Sigma)$, $\Sigma \succ 0$, and $i \neq j$:*

$$X_i \perp\!\!\!\perp X_j \mid X_{V\setminus\{i,j\}} \iff \Theta_{ij} = 0 \iff \rho_{ij\cdot rest} = 0.$$

Proof. Write $A = \{i, j\}$, $B = V \setminus A$ and, without loss of generality, $\mu = 0$. The density is $p(x) \propto \exp(-\frac{1}{2}x^\top \Theta x)$ and

$$x^\top \Theta x = \Theta_{ii}x_i^2 + \Theta_{jj}x_j^2 + 2\Theta_{ij}x_i x_j + 2x_i\,\Theta_{iB}x_B + 2x_j\,\Theta_{jB}x_B + x_B^\top \Theta_{BB} x_B.$$

Hence $p(x) = \exp(-\Theta_{ij}x_i x_j)\cdot g(x_i, x_B)\,h(x_j, x_B)$ with $g, h > 0$ collecting the remaining terms. If $\Theta_{ij} = 0$, p has the form $g(x_i, x_B)h(x_j, x_B)$ and the (density version of) Proposition 5.2 gives $X_i \perp\!\!\!\perp X_j \mid X_B$. Conversely, if $X_i \perp\!\!\!\perp X_j \mid X_B$ then $p(x) = g'(x_i, x_B)h'(x_j, x_B)$, so $\log p$ has vanishing mixed partial derivative $\partial^2 \log p / \partial x_i \partial x_j$; but from the quadratic form this derivative equals $-\Theta_{ij}$. So $\Theta_{ij} = 0$. Finally, the conditional law of X_A given X_B is Gaussian with covariance $(\Theta_{AA})^{-1}$ (Schur complement; see Section 5.14), whose correlation is $-\Theta_{ij}/\sqrt{\Theta_{ii}\Theta_{jj}}$, giving the partial-correlation form. □

Worked Example 5.13 (A Gaussian chain: dense covariance, sparse precision)**.** Let $X_1 \to X_2 \to X_3 \to X_4$ be a stationary Gaussian AR(1) chain with $\rho = 0.8$: $X_1 \sim N(0,1)$ and $X_{k+1} = \rho X_k + \sqrt{1-\rho^2}\,\varepsilon_{k+1} = 0.8X_k + 0.6\,\varepsilon_{k+1}$ with $\varepsilon_2, \varepsilon_3, \varepsilon_4 \overset{\text{iid}}{\sim} N(0,1)$ independent of X_1. *Covariance.* $\text{Var}(X_{k+1}) = 0.64 \cdot 1 + 0.36 \cdot 1 = 1$, so all variances are 1; and $\text{Cov}(X_k, X_{k+1}) = \text{Cov}(X_k, 0.8X_k + 0.6\varepsilon_{k+1}) = 0.8\text{Var}(X_k) = 0.8$ because the noise is independent of X_k; repeating

the argument, $\mathrm{Cov}(X_k, X_{k+h}) = 0.8\,\mathrm{Cov}(X_k, X_{k+h-1}) = 0.8^h$, i.e. $\Sigma_{ij} = 0.8^{|i-j|}$. So Σ is dense: $\Sigma_{13} = 0.64$, $\Sigma_{14} = 0.512$. *Precision.* Inverting Σ gives

$$\Theta = \frac{1}{0.36}\begin{pmatrix} 1 & -0.8 & 0 & 0 \\ -0.8 & 1.64 & -0.8 & 0 \\ 0 & -0.8 & 1.64 & -0.8 \\ 0 & 0 & -0.8 & 1 \end{pmatrix} = \begin{pmatrix} 2.7778 & -2.2222 & 0 & 0 \\ -2.2222 & 4.5556 & -2.2222 & 0 \\ 0 & -2.2222 & 4.5556 & -2.2222 \\ 0 & 0 & -2.2222 & 2.7778 \end{pmatrix},$$

i.e. $\Theta_{11} = \frac{1}{1-\rho^2}$, $\Theta_{22} = \frac{1+\rho^2}{1-\rho^2}$, $\Theta_{12} = \frac{-\rho}{1-\rho^2}$ with $1-\rho^2 = 0.36$. To verify, multiply row 1 of Σ by columns of the bracketed matrix and divide by 0.36: column 1 gives $(1\cdot 1 + 0.8\cdot(-0.8))/0.36 = 0.36/0.36 = 1$, column 3 gives $(0.8\cdot(-0.8) + 0.64\cdot 1.64 + 0.512\cdot(-0.8))/0.36 = (-0.64 + 1.0496 - 0.4096)/0.36 = 0$. Θ is tridiagonal: every non-neighbour pair is CI given the rest (Theorem 5.4). *Partial correlations* of neighbours: at the ends $\rho_{12\cdot 34} = 2.2222/\sqrt{2.7778 \cdot 4.5556} = 0.8/\sqrt{1.64} = 0.6247$; in the middle $\rho_{23\cdot 14} = 2.2222/4.5556 = 0.8/1.64 = 0.4878$. With only X_1, X_2, X_3 the three-variable formula gives $\rho_{13\cdot 2} = (0.64 - 0.8\cdot 0.8)/\sqrt{(1-0.8^2)(1-0.8^2)} = 0/0.36 = 0$, even though $\rho_{13} = 0.64$.

PYTHON LAB · s05_examples.py · ex_gaussian_precision()

```
Theta_11 = 2.7778, Theta_22 = 4.5556, Theta_12 = -2.2222, Theta_13 = Theta_14
= Theta_24 = 0
(Sigma Theta)_11 = 1.0000, (Sigma Theta)_13 = 0.0000   partial corr:  ends
0.6247, middle 0.4878   rho_13.2 = 0.0000
```

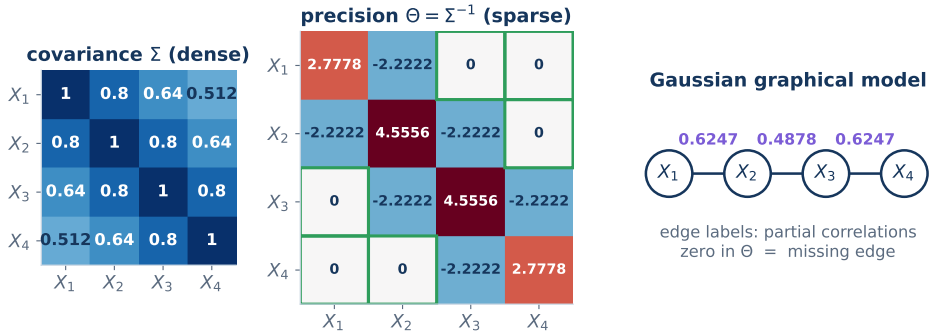


Figure 5.10. Gaussian CI for the AR(1) chain of Worked Example 5.13: the covariance $\Sigma_{ij} = 0.8^{|i-j|}$ is dense, the precision matrix Θ is tridiagonal, and the zero pattern of Θ (green boxes) is exactly the set of missing edges of the undirected Gaussian graphical model (edge labels: partial correlations given the other two variables). Session 8 builds Gaussian Markov random fields on this fact.

PART IV · STATISTICS

5.10 Sufficiency as Conditional Independence

MATHEMATICAL STATISTICS · Sufficient statistics

A statistic $T = T(X)$ is *sufficient* for a parametric family $\{f(x \mid \theta)\}$ if the conditional distribution of X given $T(X) = t$ does not depend on θ. In CI language: once T is known, the remaining detail in the data is pure noise as far as θ is concerned.

Theorem 5.5 (Fisher–Neyman factorization)**.** *T is sufficient for θ if and only if there are functions g, h with*

$$f(x \mid \theta) = g\big(T(x), \theta\big)\, h(x) \qquad \text{for all } x, \theta.$$

In the Bayesian view, where θ is a random variable with any prior π, sufficiency means $X \perp\!\!\!\perp \theta \mid T(X)$, and the posterior depends on the data only through T.

Proof. (Discrete case.) ($\Leftarrow$) For x with $T(x) = t$: $P_\theta(T{=}t) = \sum_{x':T(x')=t} g(t,\theta)h(x') = g(t,\theta)H(t)$ with $H(t) = \sum_{x':T(x')=t} h(x')$, so $P_\theta(X{=}x \mid T{=}t) = g(t,\theta)h(x)/(g(t,\theta)H(t)) = h(x)/H(t)$, free of θ. ($\Rightarrow$) If $k(x) = P(X{=}x \mid T{=}T(x))$ does not depend on θ, then $f(x \mid \theta) = P_\theta(T{=}T(x))\, k(x)$: take $g(t,\theta) = P_\theta(T{=}t)$ and $h = k$. Bayesian view: the joint of (θ, X) is $\pi(\theta)g(T(x),\theta)h(x) = [\pi(\theta)g(T(x),\theta)] \cdot [h(x)]$, a product of a function of (θ, T) and a function of (x, T) (note T is a function of x), so by Proposition 5.2 $\theta \perp\!\!\!\perp X \mid T$ and $\pi(\theta \mid x) \propto \pi(\theta)g(T(x),\theta)$. □

The parallel is exact: Proposition 5.2 says CI is a factorization $g(x,z)h(y,z)$; Fisher–Neyman says sufficiency is the factorization $g(t,\theta)h(x)$ with $Z = T$. A sufficient statistic is the "Z" that screens the data off from the parameter.

Worked Example 5.14 (Bernoulli sample)**.** For $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bern}(\theta)$, $f(x \mid \theta) = \theta^t(1{-}\theta)^{n-t}$ with $t = \sum_i x_i$: take $g(t,\theta) = \theta^t(1-\theta)^{n-t}$, $h \equiv 1$. Since $T \sim \text{Bin}(n,\theta)$, $P(T{=}t) = \binom{n}{t}\theta^t(1-\theta)^{n-t}$, and the ratio $P(x)/P(T{=}t) = 1/\binom{n}{t}$ does not contain θ. Numerically, for $x = (1,0,1,0)$ we have $n = 4$, $t = 2$, $\binom{4}{2} = 6$: with $\theta = 0.2$, $P(x) = 0.2^2 \cdot 0.8^2 = 0.04 \cdot 0.64 = 0.02560$ and $P(T{=}2) = 6 \cdot 0.02560 = 0.15360$; with $\theta = 0.7$, $P(x) = 0.7^2 \cdot 0.3^2 = 0.49 \cdot 0.09 = 0.04410$ and $P(T{=}2) = 6 \cdot 0.04410 = 0.26460$; in both cases $P(x \mid T{=}2) = 1/6 = 0.1667$. Under a uniform prior $\text{Beta}(1,1)$, the posterior is proportional to $\theta^t(1{-}\theta)^{n-t}$, i.e. $\text{Beta}(1{+}t, 1{+}n{-}t) = \text{Beta}(3,3)$, whatever the order of the observations.

PYTHON LAB · `s05_examples.py` · `ex_sufficiency_bernoulli()`

```
theta=0.2:  P(x|T)=0.02560/0.15360=0.1667   theta=0.7:
P(x|T)=0.04410/0.26460=0.1667   Beta(3,3)
```

Remark 5.1 (Exchangeability and de Finetti). A sequence $X_1, X_2, \dots$ is *exchangeable* if its law is invariant under finite permutations. De Finetti's theorem: an infinite exchangeable binary sequence is a mixture of i.i.d. sequences, i.e. $X_1, X_2, \dots \overset{\text{iid}}{\sim} \text{Bern}(\theta)$ given a random θ. Exchangeability is *exactly* "observations CI given a latent parameter" — naive Bayes with D replaced by θ, and the justification of Bayesian learning in Session 10.

PART IV · STATISTICS

5.11 Testing Conditional Independence from Data

Given an i.i.d. sample, we want to test $H_0 : X \perp\!\!\!\perp Y \mid Z$. Two classical tests cover the discrete and Gaussian cases; both are the building blocks of constraint-based structure learning (the PC algorithm, Session 11).

Discrete: the G^2 (likelihood-ratio) test. With counts N_{xyz}, margins N_{xz}, N_{yz}, N_z and $N = \sum N_{xyz}$,

$$G^2 = 2 \sum_{x,y,z} N_{xyz} \log \frac{N_{xyz} N_z}{N_{xz} N_{yz}} \overset{H_0}{\approx} \chi^2_{(|\mathcal{X}|-1)(|\mathcal{Y}|-1)|\mathcal{Z}|}.$$

Pearson's $\chi^2 = \sum (N_{xyz} - E_{xyz})^2 / E_{xyz}$ with $E_{xyz} = N_{xz} N_{yz} / N_z$ is asymptotically equivalent.

Proposition 5.5 (G^2 is twice N times the empirical CMI)**.** *$G^2 = 2N\,\hat{I}(X;Y \mid Z)$, where $\hat{I}$ is the conditional mutual information (in nats) of the empirical distribution $\hat{P} = N_{xyz}/N$. Under H_0 and regularity conditions, $G^2 \xrightarrow{d} \chi^2_{\mathrm{df}}$ with $\mathrm{df} = (|\mathcal{X}|-1)(|\mathcal{Y}|-1)|\mathcal{Z}|$.*

Proof. Substitute $\hat{P}(x,y,z) = N_{xyz}/N$ into the definition of $I(X;Y \mid Z)$: $\hat{P}(x,y \mid z)/(\hat{P}(x \mid z)\hat{P}(y \mid z)) = N_{xyz} N_z / (N_{xz} N_{yz})$, so $N\hat{I} = \sum N_{xyz} \log \frac{N_{xyz} N_z}{N_{xz} N_{yz}} = G^2/2$. For the limit: G^2 is the likelihood-ratio statistic $2(\ell_{\mathrm{sat}} - \ell_{H_0})$ comparing the saturated multinomial with the CI model, whose MLE is $\hat{P}(z)\hat{P}(x \mid z)\hat{P}(y \mid z)$. The saturated model has $|\mathcal{X}||\mathcal{Y}||\mathcal{Z}| - 1$ free parameters and the CI model $(|\mathcal{Z}|-1) + |\mathcal{Z}|(|\mathcal{X}|-1+|\mathcal{Y}|-1)$; their difference is $(|\mathcal{X}|-1)(|\mathcal{Y}|-1)|\mathcal{Z}|$, and Wilks' theorem (Casella & Berger 2002, Thm. 10.3.3) gives the χ^2 limit. □

Gaussian: the Fisher-z test. For a sample partial correlation $\hat{r} = \hat{\rho}_{ij \cdot S}$ with conditioning set S, Fisher's transform $z = \frac{1}{2} \log \frac{1+\hat{r}}{1-\hat{r}}$ is approximately $N\big(\frac{1}{2} \log \frac{1+\rho}{1-\rho},\, 1/(n - |S| - 3)\big)$, so under $H_0 : \rho_{ij \cdot S} = 0$ the statistic $\sqrt{n - |S| - 3}\,|z|$ is compared with $N(0,1)$ quantiles.

Worked Example 5.15 (Testing CI on simulated data)**.** All numbers below are *simulation outputs* of `ex_ci_tests()` (one random sample drawn with `np.random.default_rng(2024)`); another seed gives different but qualitatively identical results.

Discrete. We simulate $n = 2000$ users from the enthusiast model of Worked Example 5.6 ($P(Z{=}1) = 0.3$; X and Y drawn independently given Z with $P(X{=}1 \mid Z) = P(Y{=}1 \mid Z) = 0.9$ if $Z = 1$ and 0.1 if $Z = 0$). The counts N_{xyz} are:

	$Z=0$ ($N_z=1411$)		$Z=1$ ($N_z=589$)	
	$Y=0$	$Y=1$	$Y=0$	$Y=1$
$X=0$	1138	124	4	41
$X=1$	136	13	54	490

Marginal test of $X \perp\!\!\!\perp Y$: summing over Z gives the table $\left(\begin{smallmatrix} 1142 & 165 \\ 190 & 503 \end{smallmatrix}\right)$ (rows $X = 0,1$, columns $Y = 0,1$) with expected counts $E_{xy} = N_x N_y / N$, e.g. $E_{00} = 1307 \cdot 1332/2000 = 870.46$. Then $G^2 = 2\sum N_{xy} \log(N_{xy}/E_{xy}) = 742.63$ on $(2-1)(2-1) = 1$ df, $p = 1.6 \times 10^{-163}$: strongly rejected. *Stratified test of $X \perp\!\!\!\perp Y \mid Z$:* within each stratum the expected counts are $E_{xyz} = N_{xz} N_{yz} / N_z$, e.g. $E_{000} = 1262 \cdot 1274/1411 = 1139.47$; the stratum contributions are $G^2_{Z=0} = 0.189$ and $G^2_{Z=1} = 0.052$, so $G^2 = 0.24$ on $(2-1)(2-1) \cdot 2 = 2$ df, $p = 0.886$: no evidence against CI. Check of Proposition 5.5 for the marginal test: the empirical mutual information (CMI with $Z = \emptyset$) is $\hat{I}(X;Y) = 0.18566$ nats and $2 \cdot 2000 \cdot 0.18566 = 742.63$.

Gaussian. We draw $n = 500$ samples of the chain $X_1 \to X_2 \to X_3$ of Worked Example 5.13 ($\rho = 0.8$). The sample correlations are $\hat{r}_{12} = 0.7729$, $\hat{r}_{23} = 0.7982$, $\hat{r}_{13} = 0.6220$. Marginal test ($S = \emptyset$): $z = \frac{1}{2} \log \frac{1.6220}{0.3780} = 0.7282$, statistic $\sqrt{500-3} \cdot 0.7282 = 16.23$, $p = 2.87 \times 10^{-59}$: reject $X_1 \perp\!\!\!\perp X_3$. Conditional test ($S = \{X_2\}$): $\hat{r}_{13 \cdot 2} = \frac{0.6220 - 0.7729 \cdot 0.7982}{\sqrt{(1-0.7729^2)(1-0.7982^2)}} = \frac{0.0050}{0.3822} = 0.0132$ (computed from the unrounded correlations; the subtraction in the numerator is sensitive to rounding), $z = 0.0132$, statistic $\sqrt{500-1-3} \cdot 0.0132 = 0.29$, $p = 0.769$: no evidence against $X_1 \perp\!\!\!\perp X_3 \mid X_2$.

PYTHON LAB · s05_examples.py · ex_ci_tests()

```
stratum Z=0:  G2_z = 0.189   stratum Z=1:  G2_z = 0.052   2*2000*0.18566 =
742.63
X indep Y: G2 = 742.63, df = 1, p = 1.6e-163   X indep Y | Z: G2 = 0.24, df
= 2, p = 0.886
r_13 = 0.6220, Fisher z = 0.7282, z-stat = 16.23, p = 2.87e-59
r_13.2 = 0.0050/0.3822 = 0.0132, z-stat = sqrt(496)*0.0132 = 0.29, p = 0.769
```

PART V

Conditional Independence in AI, Agents and RL

Belief states, Markov property, routers and multi-agent fusion

PART V · CONNECTIONS

5.12 Connections to AI, Agentic AI, RL and Multi-Agent Systems

5.12.1 Agentic AI: belief states and naive-Bayes routing

AI CONNECTION · Agentic AI

An LLM assistant does not observe the user's intent U; it observes utterances $O_1, O_2, \ldots$. If utterances are modelled as CI given the intent, $O_t \perp\!\!\!\perp O_{1:t-1} \mid U$, the agent's *belief state* $b_t(u) = P(U{=}u \mid o_{1:t})$ satisfies the recursion

$$b_t(u) \;\propto\; P(o_t \mid u)\, b_{t-1}(u),$$

one multiplication per turn, with memory independent of conversation length. As a *naive-Bayes router* the agent calls the tool (billing API, troubleshooter, CRM) of the most probable intent.

Worked Example 5.16 (Tracking a hidden user intent)**.** Intents $u \in \{\text{Billing, TechSupport, Sales}\}$ with prior $b_0 = (0.5, 0.3, 0.2)$ (all vectors below are in this order); hypothetical likelihoods $P(\text{“invoice”} \mid u) = (0.6, 0.1, 0.2)$ and $P(\text{“error”} \mid u) = (0.2, 0.7, 0.05)$ for the words the user types in turns 1 and 2. Each update multiplies the current belief by the likelihood entrywise and divides by the sum (the evidence).

- After “invoice”: unnormalized $(0.5 \cdot 0.6,\ 0.3 \cdot 0.1,\ 0.2 \cdot 0.2) = (0.30, 0.03, 0.04)$, evidence $0.30 + 0.03 + 0.04 = 0.37$, belief $b_1 = (0.30, 0.03, 0.04)/0.37 = (0.8108, 0.0811, 0.1081)$.
- After “error” (CI given u lets us reuse b_1 as the prior): unnormalized $(0.8108 \cdot 0.2,\ 0.0811 \cdot 0.7,\ 0.1081 \cdot 0.05) = (0.1622, 0.0568, 0.0054)$, evidence 0.2243, belief $b_2 = (0.7229, 0.2530, 0.0241)$.

Billing still wins, but TechSupport has more than tripled ($0.0811 \to 0.2530$): a clarifying

question may be worth its cost (value of information, course session 21).

PYTHON LAB · s05_examples.py · ex_agent_intent()

```
after 'invoice':  unnormalised (0.3000, 0.0300, 0.0400), evidence 0.3700,
belief (0.8108, 0.0811, 0.1081)
after 'error':    unnormalised (0.1622, 0.0568, 0.0054), evidence 0.2243,
belief (0.7229, 0.2530, 0.0241)
```

LINK TO COURSE SESSION 26 · Agentic AI

The CI assumption $O_t \perp\!\!\!\perp O_{1:t-1} \mid U$ lets an agent's memory be a fixed-size belief vector rather than the full transcript; when it fails (users refer back to earlier turns) the latent state must be enriched — the POMDP view below.

5.12.2 Reinforcement learning: the Markov property

AI CONNECTION · Reinforcement Learning

An MDP is defined by a CI statement. With history $H_{t-1} = (S_0, A_0, \ldots, S_{t-1}, A_{t-1})$, the *Markov property* reads

$$S_{t+1} \perp\!\!\!\perp H_{t-1} \mid S_t, A_t, \qquad \text{equivalently} \qquad P(s_{t+1} \mid s_{0:t}, a_{0:t}) = P(s_{t+1} \mid s_t, a_t).$$

It makes the transition model a single table $P(s' \mid s, a)$. In a POMDP the state is hidden and observations satisfy $O_t \perp\!\!\!\perp$ (everything else) $\mid S_t$; the belief state $b_t(s) = P(S_t{=}s \mid o_{1:t}, a_{0:t-1})$ is itself Markov, with update $b_{t+1}(s') \propto P(o_{t+1} \mid s') \sum_s P(s' \mid s, a_t) b_t(s)$, which is why POMDPs reduce to MDPs over beliefs (course session 21).

Worked Example 5.17 (Testing the Markov property of a state and of an observation)**.** *Setup (simulation, `ex_markov_property()`, seed 17).* A robot moves in a corridor of 4 cells $S_t \in \{0, 1, 2, 3\}$, starting in cell 0. At each step it picks $A_t \in \{L, R\}$ uniformly at random; the move succeeds with probability 0.8 (otherwise it stays), and a move into a wall leaves it in place. We simulate 20 000 steps and test the CI statements with the stratified G^2 test of Section 5.11, one stratum per value of the conditioning variables. *True state.* For $S_{t+1} \perp\!\!\!\perp S_{t-1} \mid S_t, A_t$ there are $4 \cdot 2 = 8$ strata. Many cells are *structurally* zero (e.g. from cell 1 the robot can only reach cells $0, 1, 2$), so each stratum contributes $(\#\text{non-empty rows} - 1)(\#\text{non-empty columns} - 1)$ df: $0, 1, 2, 2, 2, 2, 1, 0$ for $(S_t, A_t) = (0, L), (0, R), \ldots, (3, R)$, in total 10 df. The test gives $G^2 = 13.60$, $p = 0.192$: consistent with Markov. *Aliased observation* $O_t = \mathbb{1}\{S_t \geq 2\}$ ("in the right half"). For $O_{t+1} \perp\!\!\!\perp O_{t-1} \mid O_t, A_t$ there are 4 strata; in $(O_t, A_t) = (0, L)$ and $(1, R)$ the next observation is certain, leaving $1 + 1 = 2$ df. The test gives $G^2 = 1684.4$, $p < 10^{-300}$ (it underflows to 0 in double precision): *not* Markov. Indeed the estimated $P(O_{t+1}{=}1 \mid O_t{=}0, A_t{=}R)$ is 0.294 if $O_{t-1} = 0$ but 0.805 if $O_{t-1} = 1$: in the second case the robot has just come from cell 2 into cell 1, so a right move reaches cell 2 with probability 0.8, whereas in the first case it may be in cell 0 or 1. Remedy: a belief over S_t or stacked observations (course sessions 20–21).

PYTHON LAB · s05_examples.py · ex_markov_property()

```
State:        G2 = 13.60, df = 10, p = 0.192   (df per stratum
0,1,2,2,2,2,1,0)
Observation:  G2 = 1684.4, df = 2, p = 0 (underflow; p < 1e-300)
P(O'=1|0,R,0) = 0.294, P(O'=1|0,R,1) = 0.805
```

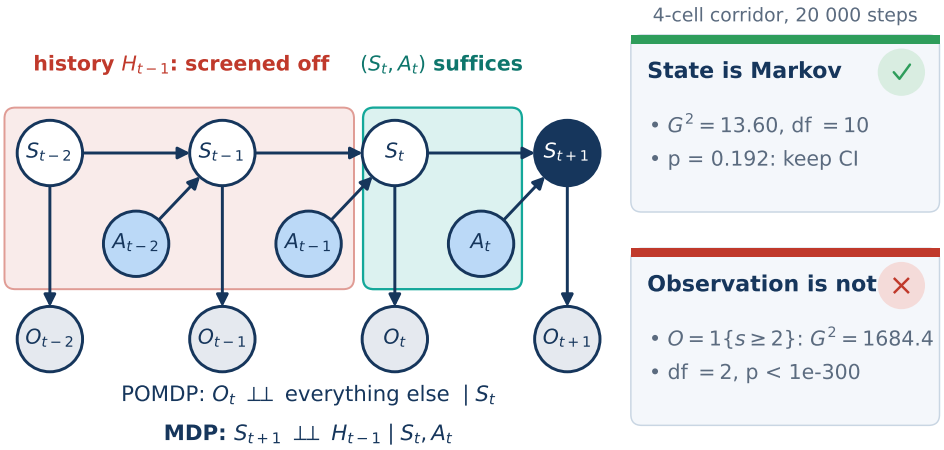


Figure 5.11. The Markov property of an MDP and the observation model of a POMDP as CI statements, with the outcome of the G^2 tests of Worked Example 5.17 (simulated 4-cell corridor, 20 000 steps).

LINK TO COURSE SESSION 17 · RL and MDPs

Policies and values are defined relative to a state with $S_{t+1}, R_{t+1} \perp\!\!\!\perp H_{t-1} \mid S_t, A_t$. The Bellman equation $V^\pi(s) = \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s, a)[r + \gamma V^\pi(s')]$ is the chain rule (5.3) along time, simplified by exactly this CI statement.

5.12.3 Multi-agent AI: private signals and sensor fusion

AI CONNECTION · Multi-Agent AI

Agents $1, \dots, m$ observe private noisy signals $Y_1, \dots, Y_m$ of a common state S (a hazard, a price, an opponent's type), assumed $Y_i \perp\!\!\!\perp Y_j \mid S$. Then the posterior log-odds is a *sum* of individual log-likelihood ratios,

$$\log \frac{P(S{=}1 \mid y_{1:m})}{P(S{=}0 \mid y_{1:m})} = \log \frac{P(S{=}1)}{P(S{=}0)} + \sum_{i=1}^{m} \log \frac{P(y_i \mid S{=}1)}{P(y_i \mid S{=}0)},$$

so each agent need only send one number. If CI fails (e.g. a relay), naive fusion double-counts evidence.

Worked Example 5.18 (Three agents and a relay). *Data (hypothetical).* Prior $P(S{=}1) = 0.3$. Agent i reports correctly with probability q_i, i.e. $P(Y_i{=}1 \mid S{=}1) = P(Y_i{=}0 \mid S{=}0) = q_i$, with $q_1, q_2, q_3 = 0.7, 0.8, 0.9$, and the reports are CI given S. *Log-likelihood ratios.* A report "1" from agent i contributes $\log \frac{q_i}{1-q_i}$ and a report "0" contributes $\log \frac{1-q_i}{q_i}$ (natural logs). The prior log-odds is $\log \frac{0.3}{0.7} = -0.8473$. *Reports* $(1, 1, 0)$ contribute $\log \frac{0.7}{0.3} = +0.8473$, $\log \frac{0.8}{0.2} = +1.3863$ and $\log \frac{0.1}{0.9} = -2.1972$, so the posterior log-odds is $-0.8473 + 0.8473 + 1.3863 - 2.1972 = -0.8109$.

In odds form, $\frac{3}{7} \cdot \frac{7}{3} \cdot 4 \cdot \frac{1}{9} = \frac{4}{9}$, hence $P(S{=}1 \mid \text{reports}) = \frac{4/9}{1+4/9} = \frac{4}{13} = 0.3077$: the most reliable agent's "no" outweighs two weaker "yes". *Reports* $(1,1,1)$: the last term becomes $+2.1972$, log-odds 3.5835, odds $\frac{3}{7} \cdot \frac{7}{3} \cdot 4 \cdot 9 = 36$, $P = \frac{36}{37} = 0.9730$. *Relay.* If a fourth agent only relays agent 2's report and is wrongly treated as an independent sensor with $q = 0.8$, the term $+1.3863$ is counted twice: log-odds $-0.8109 + 1.3863 = 0.5754$, odds $\frac{4}{9} \cdot 4 = \frac{16}{9}$, $P = \frac{16}{25} = 0.6400$ — more than double $(0.6400/0.3077 = 2.08)$ the correct 0.3077 (Figure 5.12).

PYTHON LAB · s05_examples.py · ex_sensor_fusion()

```
signals (1,1,0):  contributions -0.8473, 0.8473, 1.3863, -2.1972; log-odds
-0.8109, P(S=1|signals) = 0.3077
signals (1,1,1):  log-odds 3.5835, P = 0.9730   relayed copy of agent 2:
log-odds 0.5754, P = 0.6400
```

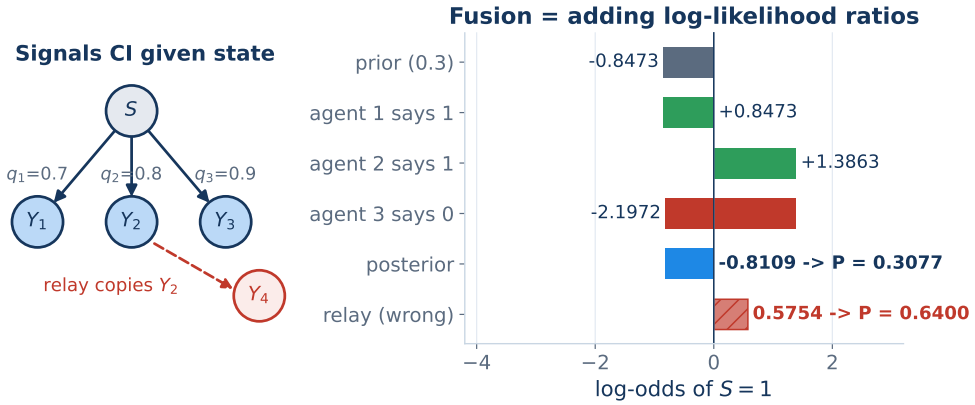


Figure 5.12. Multi-agent sensor fusion (Worked Example 5.18). Left: private signals CI given the state, with accuracies $q_i = P(\text{agent } i \text{ reports correctly})$, and a relay Y_4 that copies Y_2 and so violates CI. Right: fusion adds log-likelihood ratios to the prior log-odds; the relayed copy double-counts agent 2 and pushes the posterior from 0.3077 to 0.6400.

LINK TO COURSE SESSION 27 · Multi-Agent AI

Communication and negotiation protocols often assume CI private information; whether it holds (shared data, gossip, herding) decides whether pooled beliefs are calibrated. The same structure underlies distributed inference by message passing (Sessions 7–8).

5.12.4 Generative models and AI paradigms

LINK TO COURSE SESSION 4 · Generative Models

A VAE decoder factorizes $p_\theta(x \mid z) = \prod_j p_\theta(x_j \mid z)$: pixels (or tokens) are CI given the latent code — the naive-Bayes structure with a continuous, learned z. All pixel dependence must be carried by z (hence blur when z is too small; autoregressive decoders drop this CI assumption). Diffusion models rely on $x_{t-1} \perp\!\!\!\perp x_{0:t-2} \mid x_t$ in the forward chain; Session 10 derives the ELBO from EM. *Paradigms (course session 2):* symbolic AI encodes relevance with rules, statistical AI with graded, learnable CI statements; the graphoid axioms are a small symbolic *logic of irrelevance* about statistical facts — a neuro-symbolic bridge.

Table 5.3. This session in the Advanced AI course: where each concept reappears.

Concept of this session	Reappears as	Course
Bayes' theorem, total probability	belief update of agents, posterior inference	1, 7, 21, 26
CI and factorization	Bayesian networks, d-separation	6, 7
Gaussian CI, graphoid axioms	Gaussian MRFs, Markov properties, Kalman filter	6, 7, 8
Sufficiency, exchangeability	Bayesian learning, conjugate priors	10
CI tests (G^2, Fisher-z)	PC algorithm, causal discovery	11, 12
Markov property	MDPs, Bellman equations, Q-learning	17, 18, 20
Observation model $O_t \perp\!\!\!\perp$ rest $\mid S_t$	POMDPs and belief states	21, 24
Naive-Bayes router, belief state	tool selection and memory in LLM agents	26
CI of private signals	sensor fusion, cooperation, negotiation	27

PART V · CONNECTIONS

5.13 Summary

Table 5.4. Cheat-sheet of key formulas.

Result	Formula
Bayes / total probability	$P(x \mid y) = P(y \mid x)P(x) / \sum_{x'} P(y \mid x')P(x')$
Law of total variance	$\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X])$
CI definition	$P(x, y \mid z) = P(x \mid z)P(y \mid z)$
Factorized form	$X \perp\!\!\!\perp Y \mid Z \iff P(x, y, z) = g(x, z)h(y, z)$
DAG factorization	$P(x_{1:n}) = \prod_i P(x_i \mid \mathrm{pa}(x_i))$
Parameter count	$\sum_i (k_i - 1) \prod_{j \in \mathrm{pa}(i)} k_j$
Information	$X \perp\!\!\!\perp Y \mid Z \iff I(X; Y \mid Z) = 0$
Gaussian CI	$X_i \perp\!\!\!\perp X_j \mid X_{\text{rest}} \iff (\Sigma^{-1})_{ij} = 0$
Sufficiency	$f(x \mid \theta) = g(T(x), \theta)h(x) \iff X \perp\!\!\!\perp \theta \mid T$
G^2 test	$G^2 = 2N\hat{I}(X; Y \mid Z) \approx \chi^2_{(\lvert\mathcal{X}\rvert-1)(\lvert\mathcal{Y}\rvert-1)\lvert\mathcal{Z}\rvert}$
Markov property	$S_{t+1} \perp\!\!\!\perp H_{t-1} \mid S_t, A_t$

Table 5.5. Common pitfalls and how to avoid them.

Pitfall	Correct view
"Independent implies conditionally independent"	False: conditioning on a common effect creates dependence (XOR example).
"Conditionally independent implies independent"	False: a common cause creates marginal dependence (enthusiast example).
Using intersection with zero probabilities	Intersection needs $P > 0$; deterministic relations break it.
Zero correlation means CI	Only for jointly Gaussian variables, and only with *partial* correlation given the right set.
Non-rejection proves CI	Low power and multiple testing; treat test results as evidence, not proof.
Fusing reports as if CI	Shared sources or relays double-count evidence.

THINK ABOUT IT **Which CI statements in your favourite ML model are assumptions, and which are guaranteed by its construction?**

THINK ABOUT IT **An LLM agent's memory is a summary of the conversation. What CI statement makes a summary "sufficient" for future decisions?**

THINK ABOUT IT **In a team of agents trained on the same data, are their errors conditionally independent given the truth? What happens to ensemble voting if not?**

Further reading. Koller & Friedman (2009), *Probabilistic Graphical Models*, Ch. 2–3 (probability review, CI, I-maps). Pearl (2009), *Causality*, Ch. 1 (graphoids). Wasserman (2004), *All of Statistics*, Ch. 1–3 and 9 (probability review, sufficiency). Casella & Berger (2002), *Statistical Inference*, Ch. 6 and 10 (sufficiency, likelihood-ratio tests). Murphy (2022), *Probabilistic Machine Learning: An Introduction* (Gaussians, precision matrices). Sutton & Barto (2018), Ch. 3 (Markov property). Russell & Norvig (2021), *AIMA* 4e.

APPENDIX

5.14 Appendix: Derivations Omitted Above

Proof of Proposition 5.2, full algebra. For the forward direction, from Definition 5.2, $P(x, y, z) = P(z)P(x, y \mid z) = P(z)P(x \mid z)P(y \mid z)$. Group as $\big[P(x \mid z)P(z)\big] \cdot \big[P(y \mid z)\big] = g(x, z)h(y, z)$. For the converse, if $P(x, y, z) = g(x, z)h(y, z)$, marginalizing over x gives $P(y, z) = h(y, z)G(z)$ where $G(z) = \sum_x g(x, z)$, and marginalizing over y gives $P(x, z) = g(x, z)H(z)$ where $H(z) = \sum_y h(y, z)$. Marginalizing over both gives $P(z) = G(z)H(z)$. Then

$$P(x, y \mid z) = \frac{g(x, z)h(y, z)}{G(z)H(z)} = \frac{g(x, z)}{G(z)} \cdot \frac{h(y, z)}{H(z)} = P(x \mid z)P(y \mid z),$$

using $P(x \mid z) = P(x, z)/P(z) = g(x, z)/G(z)$ and similarly for $P(y \mid z)$.

Proof of symmetry, decomposition, weak union. Symmetry is immediate from Definition 5.2 since multiplication commutes. For decomposition, sum both sides of $P(x, y, w \mid z) = P(x \mid z)P(y, w \mid z)$ over w to get $P(x, y \mid z) = P(x \mid z)P(y \mid z)$. For weak union, start again from $P(x, y, w \mid z) = P(x \mid z)P(y, w \mid z)$ and divide both sides by $P(w \mid z) > 0$: the left side becomes $P(x, y \mid z, w)$ and the right side $P(x \mid z)P(y \mid z, w)$. Summing this identity over y gives $P(x \mid z, w) = P(x \mid z)$; substituting back yields $P(x, y \mid z, w) = P(x \mid z, w)P(y \mid z, w)$, i.e. $X \perp\!\!\!\perp Y \mid (Z, W)$. (Students may find it faster to verify weak union graphically in Session 6, where it becomes a one-line fact about separation.)

Proof of intersection (Proposition 5.4, (v)). Assume $P > 0$. The hypotheses give $P(x \mid y, w, z) = P(x \mid w, z)$ and $P(x \mid y, w, z) = P(x \mid y, z)$, hence $P(x \mid w, z) = P(x \mid y, z)$ for all x, y, w, z. The left side does not depend on y, the right side not on w; since all combinations (y, w) occur, both equal a function $f(x, z)$. Summing $P(x \mid y, z)P(y \mid z) = f(x, z)P(y \mid z)$ over y shows $f(x, z) = P(x \mid z)$. Therefore $P(x \mid y, w, z) = P(x \mid z)$, i.e. $X \perp\!\!\!\perp (Y, W) \mid Z$. Worked Example 5.3 shows the step "all combinations occur" cannot be dropped.

Proof of Theorem 5.2, ($\Leftarrow$). Suppose (5.5) holds and fix X_i. Let $D = \mathrm{de}(X_i)$ and $N = \mathrm{nd}(X_i)$. Summing (5.5) over the values of D, eliminating a descendant that has no remaining children at each step (each such CPT sums to one), leaves $P(x_i, x_N) = P(x_i \mid \mathrm{pa}(x_i)) \prod_{j \in N} P(x_j \mid \mathrm{pa}(x_j))$, since parents of non-descendants are non-descendants. The first factor is a function of $(x_i, \mathrm{pa}(x_i))$ and the product a function of $x_N \supseteq \mathrm{pa}(x_i)$; by Proposition 5.2 with $Z = \mathrm{pa}(X_i)$, $X_i \perp\!\!\!\perp N \setminus \mathrm{pa}(X_i) \mid \mathrm{pa}(X_i)$.

Uniqueness of the smallest parent set under positivity (used in the I-map subsection of Section 5.7). Fix $X = X_i$ and its predecessor set Pred, and suppose $U_1, U_2 \subseteq \mathrm{Pred}$ both satisfy (5.4). Write $A = U_1 \cap U_2$, $B = U_1 \setminus U_2$, $C = U_2 \setminus U_1$, $R = \mathrm{Pred} \setminus (U_1 \cup U_2)$, so the hypotheses read $X \perp\!\!\!\perp (C, R) \mid (A, B)$ and $X \perp\!\!\!\perp (B, R) \mid (A, C)$. (1) Weak union gives $X \perp\!\!\!\perp C \mid (A, B, R)$ and $X \perp\!\!\!\perp B \mid (A, C, R)$; (2) intersection (needs $P > 0$) gives $X \perp\!\!\!\perp (B, C) \mid (A, R)$, and decomposition gives $X \perp\!\!\!\perp B \mid (A, R)$; (3) decomposition of the first hypothesis gives $X \perp\!\!\!\perp R \mid (A, B)$, and intersection with $X \perp\!\!\!\perp B \mid (A, R)$ gives $X \perp\!\!\!\perp (B, R) \mid A$, hence $X \perp\!\!\!\perp R \mid A$; (4) contraction of $X \perp\!\!\!\perp R \mid A$ and $X \perp\!\!\!\perp (B, C) \mid (A, R)$ gives $X \perp\!\!\!\perp (B, C, R) \mid A$, i.e. $A = U_1 \cap U_2$ also satisfies (5.4). If $U_1 \neq U_2$ had the same minimal size, A would be strictly smaller — a contradiction.

Gaussian conditioning (used in Theorem 5.4). Partition Θ into blocks A, B. Completing the square in x_A, $x^\top \Theta x = (x_A + \Theta_{AA}^{-1}\Theta_{AB}x_B)^\top \Theta_{AA}(x_A + \Theta_{AA}^{-1}\Theta_{AB}x_B) + x_B^\top(\Theta_{BB} - \Theta_{BA}\Theta_{AA}^{-1}\Theta_{AB})x_B$, so $X_A \mid X_B = x_B \sim N(-\Theta_{AA}^{-1}\Theta_{AB}x_B,\ \Theta_{AA}^{-1})$. For $A = \{i, j\}$, $\Theta_{AA}^{-1} = \frac{1}{\Theta_{ii}\Theta_{jj} - \Theta_{ij}^2}\begin{pmatrix} \Theta_{jj} & -\Theta_{ij} \\ -\Theta_{ij} & \Theta_{ii} \end{pmatrix}$, whose correlation is $-\Theta_{ij}/\sqrt{\Theta_{ii}\Theta_{jj}}$.

CONNECTIONS AND TAKEAWAYS

This session established that (1) the chain rule always factors a joint distribution, but only conditional independence makes the factors small; (2) $X \perp\!\!\!\perp Y \mid Z$ has an equivalent product-factorization form (Proposition 5.2) that foreshadows how we will compute with factors rather than full tables; (3) CI obeys algebraic rules (symmetry, decomposition, weak union, contraction, and intersection under positivity) that let us reason about many independence statements at once; and (4) CI can be measured, tested, and recognized in agents, RL and multi-agent systems.

What we do *not* yet have is a way to specify or read off *which* conditional independencies hold among n variables without writing out a wall of $\perp\!\!\!\perp$ statements by hand. **Session 6** answers this with a single idea: draw a directed graph whose missing edges encode conditional

independence assumptions, and prove that a simple graphical test (d-separation) tells you exactly which CI statements the graph implies — turning today's algebra into a picture you can read at a glance.